DIVERSEMIND: AN INTEGRATED FRAMEWORK FOR CHILDREN WITH MULTI-DIMENSIONAL CHALLENGES AS SLOW LEARNERS

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Dissertation submitted in partial fulfilment of the requirements for the Bachelor of Science Special Honors Degree in Information Technology

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DECLARATION

I declare that this is my own work, and this dissertation does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text. Also, I hereby grant to Sri Lanka Institute of Information Technology the non-exclusive right to reproduce and distribute my dissertation in whole or part in print, electronic or other medium. I retain the right to use this content in whole or part in future works (such as articles or books).

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Signature of Co-Supervisor	Date

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ABSTRACT

Education systems worldwide face challenges in supporting slow learners, who often struggle in traditional classrooms due to difficulties in different areas of learning. However, existing educational frameworks in Sri Lanka lack comprehensive screening and targeted interventions, limiting the potential of slow learners. To address this, an integrated framework “DiverseMind” has been introduced for Grade 4 primary school children, utilizing advanced machine learning and deep learning algorithms to assess writing skills, mathematical proficiency, attention span, and short-term memory. A key focus of the framework is evaluating attention span, a critical yet often overlooked factor in academic performance. Using a multi-model approach incorporating facial landmark detection, gaze tracking, blink count detection, and emotion recognition, a Convolutional Neural Network (CNN) model trained on 28,709 images and tested on 7,178 images achieved an accuracy of 78.12%, enabling real-time attention evaluation. The system features kid-friendly interfaces with lively animations to keep students focused and engaged, while providing instant feedback. Teachers can monitor detailed attention results through a dedicated dashboard. Gamification was introduced as an intervention to strengthen attention skills. Therefore, the solution incorporates gamified activities designed to enhance focus and cognitive abilities by promoting selective and sustained attention, cognitive control, and divided attention. The system was tested in schools, and attention evaluations showed promising results consistent with expectations. The proposed solution, which includes both screening and intervention, was highly recommended by school teachers and principals. It was particularly valued for its gamified approach, which not only supports children's improvement but also ensures a more enjoyable and engaging learning experience, making it well-suited for wider adoption in primary education. Therefore, this research bridges the educational support gap for slow learners in Sri Lanka, offering an inclusive, engaging, and supportive learning environment.

Keywords - Slow Learner, Attention Span, Convolutional Neural Networks, Sri Lanka

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LIST OF ABBREVIATIONS

Abbreviation	**Description**
ADHD	Attention-Deficit and Hyperactivity Disorder
API	Application Programming Interface
BCI	Brain Computer Interaction
CNN	Convolutional Neural Networks
CPAM	Coordinate Pair Angle Method
DNN	Deep Neural Networks
DT	Decision Tree
HPE	Head Posture Estimation
IDE	Integrated Development Environment
KNN	K-Nearest Neighbor
LD	Learning Disability
ML	Machine Learning
RF	Random Forest
SDLC	Software Development Life Cycle
SVM	Support Vector Machine
UI	User Interface

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1. INTRODUCTION

1.1 Background

In under-developing countries like Sri Lanka, "Slow Learners" often remain overlooked within the education system, receiving minimal support to realize their full potential. According to the study [1], slow learners are described as average students who may not respond well to traditional teaching methods. It also highlights that many such learners have the potential to become successful professionals, such as scientists, doctors, engineers, or writers. However, these students encounter numerous obstacles throughout their educational journey. Failing to recognize their learning difficulties early on, along with the lack of timely support, may significantly limit their progress. Early identification can offer these students a greater opportunity to develop their abilities, allowing more time and resources to support their growth. The case study [2] characterizes a slow learner as someone performing below average and emphasizes the difficulties educators face in managing such students in the classroom. It further indicates that these learners are often perceived as weak and are at risk of dropping out, which can leave lasting negative impacts on their lives. Table 1.1 outlines the socioeconomic factors that contribute to slow learning.

Table 1.1:Socioeconomic factors that influence slow learners

Independent variables	**'r' value**
Ordinal position	0.16 NS
Size of the family	0.09 NS
Parents' education - fathers' education, mothers' education	0.39*,0.30*
Parents' occupation- fathers' occupation, mothers' occupation	0.25*,0.22*
Per capita income	0.24*

*Significant at 0.05 per cent level, NS – Non-significant

Source: Socioeconomic and educational study of slow learners, 2013

As per the analysis [3], illustrated in Table 1.1, the ordinal position of a child and the size of the family do not influence the learning pace of a slow learner. However, factors such as parents' level of education, their occupation, and the household's per capita income do play a significant role in affecting the child's learning rate. Although these elements are tied to the child's family background and cannot be directly controlled, there are still ways to ease the challenges faced by slow learners through early identification and appropriate intervention. The research [4] shows that the detection of slow learners is not limited to academic factors. It also considers other aspects where teachers can offer individual support to the students. Furthermore, the study shows that learners can be identified using classification methods like Naïve Bayes, J48, ZeroR, and Random Tree, all implemented through the open-source tool WEKA. Korikana [5] identifies several factors influencing student education and highlights key elements specifically affecting slow learners, as represented in Figure 1.1.

Figure 1.1: Factors affecting slow learners

Source: AITU scientific research journal, 2023

The study [5], as illustrated in Figure 1.1, emphasizes the importance of recognizing all the factors that influence slow learners in order to offer them proper support and intervention. The analysis [6] discusses how education plays a vital role in achieving success both personally and within society. It also highlights the difficulty in identifying slow learners within the educational framework. According to the data [7], learning methods in the 21st century have evolved due to advancements in technology, making its integration into education essential. Interactive games have emerged as a widely used and effective approach in teaching and learning processes. The findings [8] indicate that gamification contributes positively to the learning experience. Since slow learners often have limited cognitive capabilities, specific theories are considered when designing gamified interfaces suitable for their needs. The exposition [9] discusses the limitations of the "one-size-fits-all" strategy, where e-learning tools are created without addressing the individual differences among students.

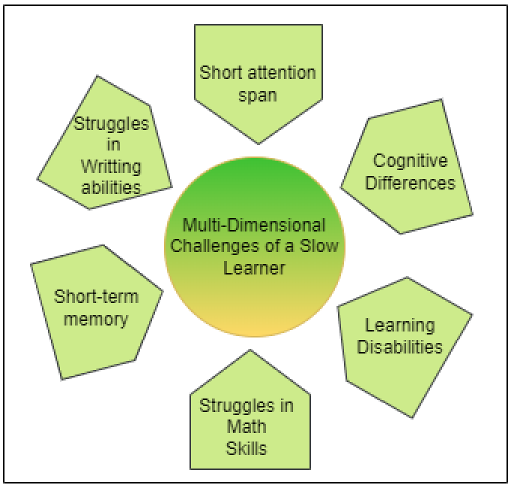


Figure 1.2: Multi-dimensional challenges faced by slow learners

Figure 1.2 illustrates the different areas that represent the multi-dimensional challenges encountered by slow learners. As a result, several key factors must be considered when addressing these difficulties. Based on this understanding, a method is proposed for the early detection of slow learners along with appropriate interventions. In alignment with this goal, “DiverseMind” concentrates on specific areas. The study [10] explains that slow learners may also have learning disabilities, which makes it essential to give them special attention. “DiverseMind” aims to assess writing skills, mathematical skills, attention span and short-term memory which applies gamified techniques to deliver meaningful and engaging support tailored to their specific learning needs.

The most common challenge faced by slow learners is the short attention span. This characteristic is prevalent among school children, and some may even exhibit symptoms of Attention-Deficit and Hyperactivity Disorder (ADHD). ADHD is usually classified based on the presence of attention problems and hyperactivity; its symptoms usually manifest before age seven [11]. The research [12] amplifies how attention predicts changes in academic skill of early childhood. Identifying the attention level of these children is crucial, as it allows for early intervention, helping them receive the support they need at an early stage.

A study explains the difficulties of sustaining attention mainly in slow learners [13]. According to reference [14], attention is fundamentally rooted in concentration and awareness. The research [15] explains about the causes of slow learners and attention span levels. This study reveals that it is because the attention span of slow learners is short, making them incapable of focusing on something for a long period of time. It is also connected with their malfunctioning short-term memory, which makes it hard for them to retain what they have learned.

The literature review identifies slow learners as inattentive and impulsive, always found to be impatient and having the tendency to blurt out answers before the question is asked and even interrupting conversations without even listening [16]. The impulsive character reflected in them is their carelessness to the extent of being inattentive and overlooking minor details. They get easily distracted and avoid tasks that require them to undertake sustained mental effort. In most cases, they seldom listen when spoken to and cannot keep their focus on a particular task.

The study [17] evaluates the attention span of learners in an online course by using a Microsoft Kinect device to capture head pose data, which is then analyzed to detect attention levels. The reference to [18] exhibits how attention level is calculated through head rotation and eye gaze direction (see Figure 1.3 and Figure 1.4).

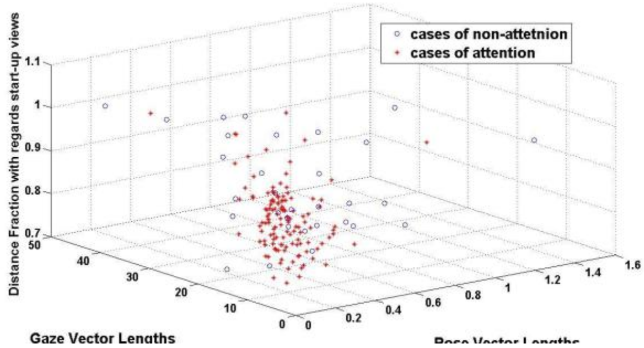


Figure 1.3: A cluster formed on cases of attentiveness and non-attentiveness

Source -The importance of eye gaze and head pose to estimating levels of attention

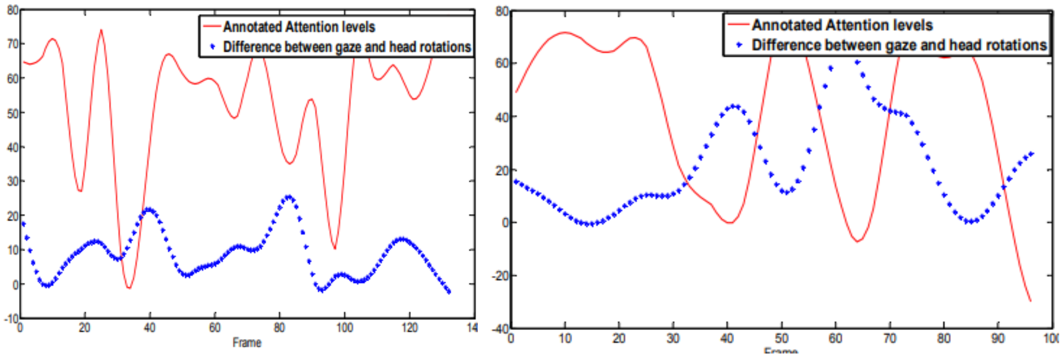


Figure 1.4: Annotation on attention levels of different gaze and head rotations

Source - The importance of eye gaze and head pose to estimating levels of attention

Effective treatment methods in patients with a short attention span were discussed in [19], especially with the help of game-based environments that would provide better focusing. A model evolving around serious games specifically crafted for the treatment of ADHD is introduced in the research. The ground study of [20] delves into emphasizing gamification methods to increase the attention of students with learning disability where gamification makes the students more interactive in the learning process more than any other methods. It also demonstrates the value of motivation in improving student's attention when utilizing educational technology and highlights the student's interest and improvement of attention levels through this gamification.

Consequently, the research "DiverseMind", emphasizes usage of Convolutional Neural Networks (CNN) based techniques where attention level of a child is evaluated focusing on slow learners. As attention level of a child significantly impacts the learning capacity, an intervention is brought for the development process of the child through gamification techniques using the Mayer's cognitive theory of Multimedia Learning. Throughout this study, CNN and Machine Learning (ML) algorithms are used for attention span evaluation. Therefore, we propose this system that calculates the attention score during computer-based tasks and generates real-time feedback calculating the overall attention span of the child using the parameters of face landmark recognition, head pose and eye gaze direction which makes it reliable to assess the student's attention levels.

1.2 Literature Survey

The ability to maintain attention is a common challenge for slow learners, often linked to their short attention span. As noted in [21], inattention is a symptom exhibited among preschoolers and not a disorder. Several studies have explored to detect and enhance attention spans with technological advancements.

In recent years, various studies have been conducted on the development of intelligent learning systems tailored for students on measuring their attention levels in online learning. The research by [22] focuses on continuously tracking the attention of students during online education. The attentiveness of the student is then computed by making use of the rate of blinking, drowsiness, eye gaze tracking, classifying emotion, identifying the face involved, making a posture check, and recognizing the noise. The research problem addressed in the study focuses on students losing interest while attending web-based lectures after a certain period of time, particularly on platforms like EdX and Coursera. This study also discusses about the attention span of ADHD students where they have issues on their concentration and attention span and recommends active learning methods as an intervention for both ADHD and other students.

Refer to [23], this research is done with the purpose of detecting the attention span of students in online learning platforms after the Covid-19 pandemic. These computer detection and feature detection methods are used to provide precise information about the student's activity. The factors of face detection, eye pose estimation, head pose estimation and drowsiness of the student determine their attention level. The research problem addressed here is the distraction of students during online learning and to help teachers check on it and bring out various teaching methods to improve the students focus. The system monitors the attention span through various factors as mentioned above and here the feature detection analyses and identifies the current movements of the face, eyes, body and determines the behavior exhibited. Deep learning is used for this process. Thus [22],[23] measures the attention of students in online education but lacks in providing feedback and targeted interventions beyond active learning recommendations.

Similarly, [24] proposes a web application that detects attention using head posture, since it gives the overall gaze direction. It also delivers feedback to both students and teachers regarding the student's level of attention. This graphs the periods of a student's best and worst attention during the lecture, along with respective analyses of such states of attention. Here it motivates students to study and helps the teacher in recognizing student behavior. However, this study focuses on primarily tracking physical attributes and not on emotion states of the student and only targets on general students. While this study highlights the importance of feedback mechanisms, it lacks emotional state recognition and does not cater specifically to slow learners, limiting its applicability to diverse learning needs.

In contrast, [25] also introduces an interactive system that focuses on measuring the attention level of learners through the features of body posture and head pose. A set test subjects were given to the participants here and the body movements and positioning were recorded using Microsoft Kinect. These test substances given to the participants in the test were assessed with the aid of three continuous performance tasks, which each lasted 1 hour. These tests went on for 14, 24, and 17 minutes respectively. An audio recorder, joint recorder, and gaze recorder were used in accomplishing this task. Likewise, in [26], the attention is estimated through head pose, for which Coordinate Pair Angle Method (CPAM) and Deep Neural Networks (DNN) are used. This was implemented for the purpose of attention span of users in all spheres of education, medical advertising and marketing. In education it is used in detecting mal-practices in e-learning based environments. This module proposes a model for mapping the user's attention using a robust regression-based model combined with CPAM and Head Posture Estimation (HPE). This module makes use of 68 facial landmark locations, an estimated head poses through angular differences, and a mapping of the user's concentration level over some time frame, producing an attention score. The ElasticNet regression model used, the CPAM model, and sequential ElasticNet regression model were implemented. Thus, a sequential regression model is customized using HPE. DNN was used to identify the underlying patterns. Here [25],[26] focuses on static assessments and does not focus on addressing the aspects by providing interventions or real-time feedback.

“BLISS” mobile application-based robot, focuses on the development of an attention-detection model with regard to child behavior [27]. In the process, it uses classifiers such as K-Nearest Neighbor (KNN), Support Vector Machine (SVM), Decision Tree (DT), Random Forest (RF), and reaches an accuracy of 70%. The model was put to work in an adaptive interaction system wherein the robot chooses actions as a result of the attention span of the child.

A different approach was taken in [28], which incorporated a gamified learning environment to improve engagement. Unlike traditional e-learning models, this study emphasized motivation and active participation. The study here proposes serious games, which uses multiple intelligence theory for the improvement of students with Learning Disability (LD) and ADHD pertaining to their research problem being that the traditional methods did not effectively engage these students and their cognitive needs to maintain their attention level. The series games used here are Boogies Academy designed for young participants of age 6 to 10 and cuibrain designed for youngsters between the age of 6 to 10. Here both of these games improve the attention span as described by Gardner’s theory of multiple intelligence. The Tree of Intelligence method is a methodology that uses the Gardner’s theory of multiple intelligence. It is used for the detection of weaker areas in one’s attention span. The standard test of WISC-IV (Wechsler Intelligence Scale for Children – Fourth Edition), D2 Attention Test and EDAH Scale are assessed for this purpose.

The research [29] uses Brain Computer Interaction (BCI) technology highlighting the potential for estimating the mental state and supporting the attention in children as the existing methods are ineffective, simple and lack reliable methods of generation. Thus, the study proposes a BCI training to enhance children's attention. To improve the engagement level the system consists of a gamification design principle. A training game designed here is developed on Unity, the Integrated Development Environment (IDE) for creating interactive media, such as real-time 2D and 3D contents, with Unity User Interface (UI) components and a game development platform. For this purpose, performance indicators and facial expressions are to be used. The design of the game will follow the attention network theory, which will have three kinds of games corresponding to the alerting, orienting, and executive control networks. Improvement in attention is achieved by repeating game tasks aligned with these attention networks.

The study emphasizes the process of using gamification to make the process of learning more engaging and interesting, aiming ADHD children who find it difficult to focus and concentrate especially in learning. The research proposed six parts of the game in gamification with the logic game for children with ADHD which are Object game, Content and Player, Design and Deploy and Play and Finish Framework [30].

The research [31] priorities on assessing the attention and level of concentration of a slow learner using game-based motion detection to aid in improving their focus while learning mathematics. Students diagnosed with LD, autism, ADHD, and impulsive behaviors, all categorized under a slow learner, are targeted in this study.

However [28], [29], [30], [31] prioritizes on providing interventions through the gamified approach it lacks the screening process on identifying the deficit. While various methodologies exist, a significant comparison emerges between Artificial Intelligence (AI) based assessment models and gamified approaches. Studies such as [22], [23], [24], [25], [26], [27] focus on predictive models to analyze student attention patterns and suggest interventions or provide feedback. Meanwhile, the studies [28], [29], [30], [31] demonstrate that gamified learning environments that leads to higher retention rates and improved engagement. The main limitation of AI-based models like [26] and [29] is their reliance on historical data, which can sometimes fail to capture real-time learning challenges, whereas [31] benefits from immediate feedback but requires careful game design to avoid distractions. Despite the advancements in personalized learning systems, several limitations persist. Existing systems like those in [31] often require significant computational power, making them less accessible in low-resource settings. Furthermore, while predication and gamification integration, combining both approaches remains underexplored.

In summary, the literature reveals diverse strategies for supporting slow learners, ranging from AI-driven assessment models to gamified learning environments. Studies such as [22]-[31] have significantly contributed to the field, but gaps remain in balancing computational efficiency, engagement, and real-time adaptability.

Thus, “DiverseMind” endeavors focusing on bridging the critical gaps by providing screening and intervention by addresses a range of unmet needs and challenges associated with supporting slow learners focusing on attention span.

1.3 Background Survey

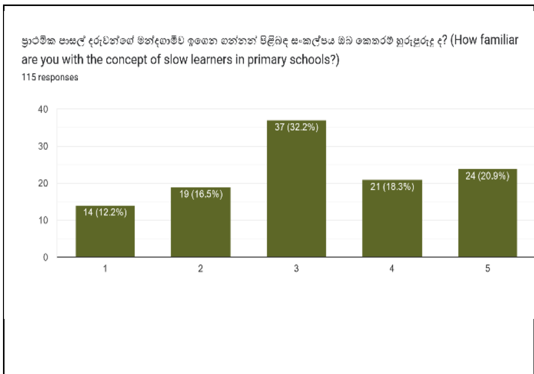
To explore the level of awareness and the importance of creating a web - based solution for identifying and assessing slow learners in primary schools, two customized questionnaires were prepared and distributed to two target groups, the general public (see Figure 1.5) and primary school teachers within the Sri Jayewardenepura Education Zone (see Figure 1.6). These surveys aimed to assess participants’ understanding, experiences, and perspectives regarding slow learners and the existing identification practices.

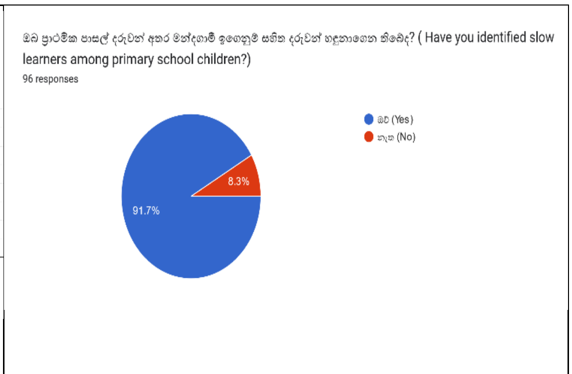
The key findings of the survey are as follows,

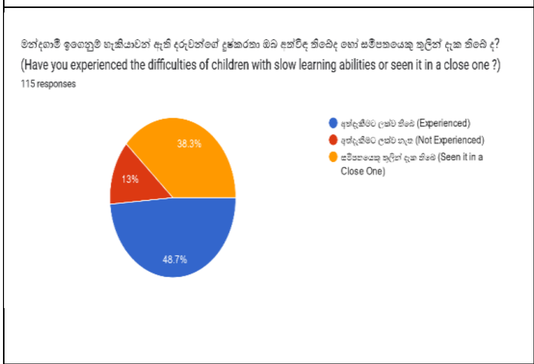
- While the general public shows limited awareness of slow learners among primary school children, teachers demonstrate significantly higher awareness.
- Both teachers and the general public reported frequent encounters with children facing learning difficulties, highlighting the prevalence and the impact of the problem.
- Participants from both groups emphasized the critical need for early identification of slow learners in primary schools.

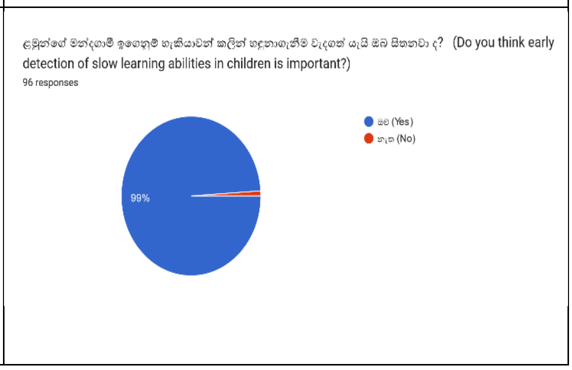
- Traditional approaches to identifying slow learners were widely considered as ineffective by respondents.
- The majority of teachers expressed interest in combining technology with conventional methods to improve the identification and support of slow learners.
- There was strong enthusiasm among both teachers and the general public for using a web application to detect and assist slow learners, recognizing its potential benefits over traditional techniques.
- A majority of participants from both groups indicated they would recommend such a web application for identifying and supporting slow learners in primary schools.

These findings show the gaps in awareness and the effectiveness of current methods, and they confirm strong support for developing a web application to address this issue.









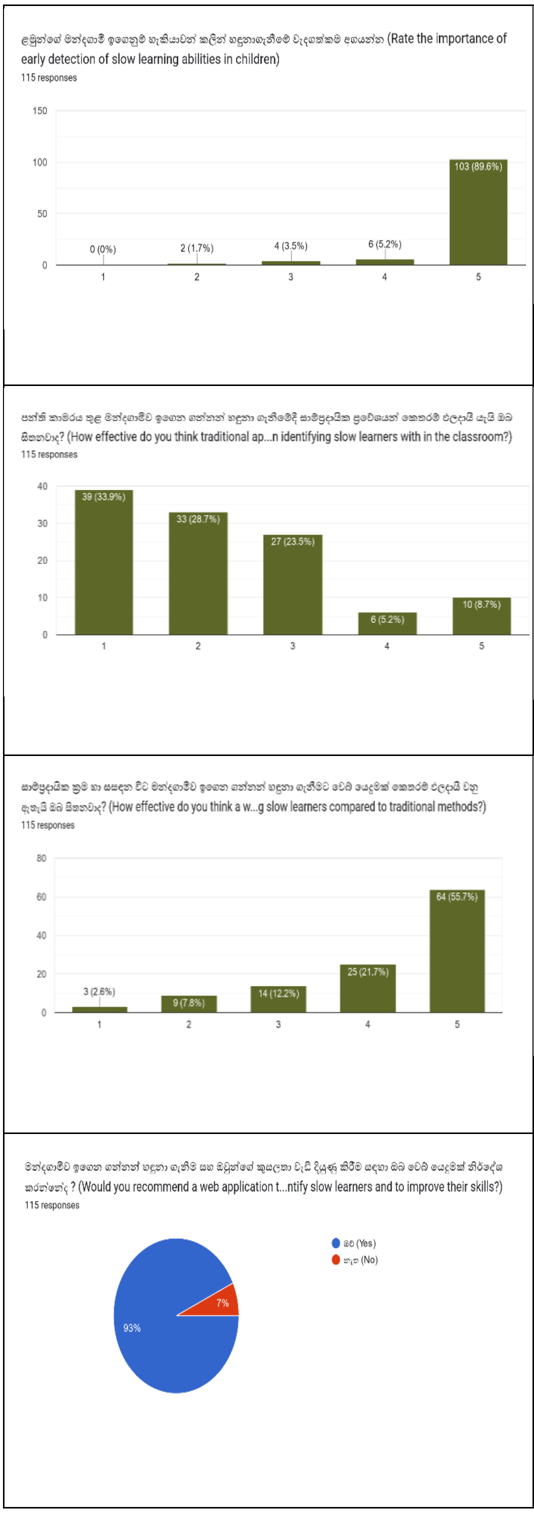


Figure 1.5: Survey result of public

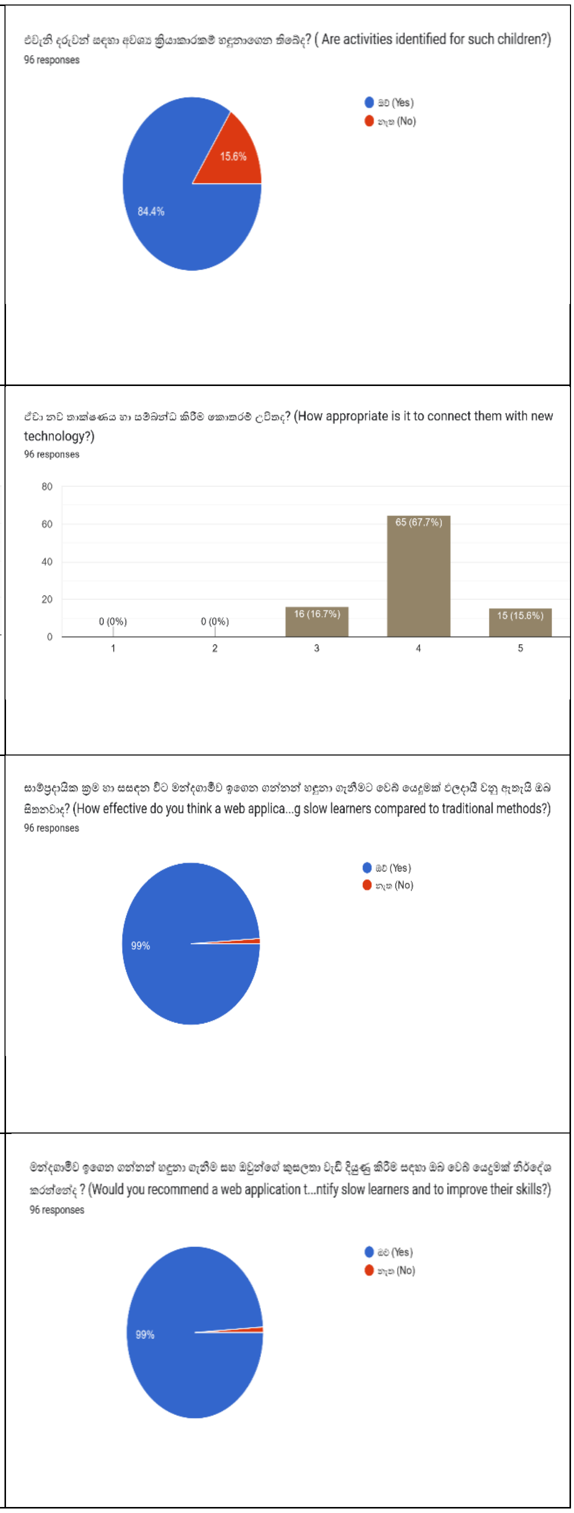


Figure 1.6: Survey result of teachers

1.4 Research Gap

The educational landscape has dramatically changed, and great interest has been devoted to understanding and optimizing the attention span of learners in various contexts. However, most of the literature works and existing systems are focused either on the general student population or on special groups, like students with ADHD or LD, often missing the unique requirements of slow learners. This is a critical gap in the literature and technology solutions available at this point in time, especially in studies that deal with slow learners across different educational settings.

While there is considerable amount of research done in the detection of attention span, assessment, and enhancement of concentration in online education, a significant gap remains concerning slow learners. The existing contributions, [22], [24], [25] pertain to attention span detection with the aid of technologies like facial landmark detection, eye-tracking systems, and body posture analysis. All of these systems are designed for broad online learner populations and do not take into consideration the special needs of slow learners. Specifically, they lack mechanisms to provide focused interventions to improve the attention level through gamified or interactive approaches. This identifies a key gap in developing a system that monitors and proactively attempts to improve the attention span of a slow learner.

The research [28] discusses using serious games to enhance the attention of LD students, it fails to incorporate attention span tracking with real-time interventions tailored for slow learners. But the absence of such a real-time adaptive learning environment that would have made dynamic response based on the slow learners represents a wide gap that exists in the research. Therefore, it is rather evident that systems need to adapt in near time to provide instant, gamified interventions ensuring that they slowly receive the support necessary to maintain and improve their focus.

Additionally, studies like [26] that employ attention prediction using head pose estimation coupled with DNN, predominantly targets general education without incorporating engagement strategies, such as gamification, that are essential for maintaining or improving slow learners' attention. This gap is enhanced by the lack of multimodal architectures that combine these detection techniques with interactive tools. Thus, it is only a methodology that embeds attention detection within a framework of engaging yet adaptive interventions that can handle the needs of slow learners.

In this light, studies [25], [31] exploit Kinect-based systems and game-based learning environments designed to measure and improve attention. But the major issue is that these systems do not have a relevant feedback mechanism or an assessment tool that is unique to slow learning. Most of these tools are very general, or they focus on just one aspect of attention, thus failing to give an integrated solution of assessment, monitoring, and tailored intervention strategy for slow learners.

This research gap is even more significant with respect to Sri Lankan education. Despite the peculiar challenges slow learners face within a typically resource-constrained educational environment, there has been minimal research into their conditions, or the formulation of specific solutions aimed at accommodating these students' needs. Traditional classroom environments in Sri Lanka frequently lack support for the diverse learning requirements of slow learners.

Finally, while current research has made progress at detection and measurement of attention spans, there remains an identified gap in the development of integrated, adaptive systems that actively seek to improve the attention span and related learning outcomes of slow learners. This is of a need within Sri Lanka, whereby desperately needed are innovative and creative solutions that can be put into use and tailored by applying gamified interventions catering to the unique challenges for slow learners within resource-constrained educational settings.

The illustrated Table 1.2 presents a comparative analysis of the existing systems to highlight the research gap.

Table 1.2: Comparison of existing systems

Features	[21]	[22]	[24]	[25]	"Diverse Mind"
Providing intervention focused on slow learners	✗	✗	✗	✗	✓
Attention enhancement gamification	✗	✗	✗	✗	✓
Aiming students	✗	✓	✗	✗	✓
Multi-Model Architecture	✓	✓	✗	✗	✓
Report Generation and feedback generation	✓	✓	✗	✗	✓
Assessment based tests	✓	✗	✓	✗	✓
Tailored based test	✗	✗	✗	✗	✓

22 - Attention Calculator for Assessment of Student and Concentration Level

21 - Real-time Attention Span Tracking in Online Education

24 - Kinect-based Dynamic Head Pose Recognition in Online Courses

25 - Attention Span Prediction using Estimation with Deep Neural Networks

1.5 Research Problem

Addressing the significant challenges faced by slow learners, particularly in developing countries like Sri Lanka, these students normally don’t receive enough support or either are not identified in traditional school systems. Refer to [32], the system of learning is designed based on the belief that children across the world undergo similar mental growth, with equal intelligence and cognitive development. Not everyone possesses the same potential and intelligence, especially slow learners. This lack of recognition leads to significant challenges, especially regarding their attention span, which is pivotal for academic success and overall development. The absence of tailored technological interventions for these students magnifies the problem, as existing educational methods fail to address their specific needs.

Slow learners struggle with various challenges, one of the most common and perhaps impactful traits is having a short attention span [1]. In a nutshell, it is the importance of the study that may potentially fill the gap, hence contributing to their academic success and holistic development. A key research problem identified is the failure to recognize that a short attention span is a primary factor contributing to their slow learning. The proposed system is designed to focus on the early identification of attention deficits with the purpose of helping to reduce the negative consequences of inattentiveness in class, avoiding poor academic performance and demotivation that eventually makes them lose confidence and motivation and drop out from school [33].The research problem identified suggests a lack of focus on tailored interventions for slow learners, particularly those that address the diverse challenges they face, such as attention deficits, in a comprehensive manner through technological approaches it could make a significant difference of slow learners in Sri Lanka and similar contexts.

The research problem identified is that most existing systems either focus on identifying attention deficits or solely on providing interventions [22]-[27], but they fail to effectively integrate both aspects into a cohesive solution. In addition, the systems created to detect attention are more costly, have less of an engaging design, and tend to discourage a student's activity and potentials without being engaging and motivational, children are unlikely to participate actively in their own development, which is important for improvement to be sustained. Many of these systems do not also bring on board expert medical advice in their design, which further limits their potential for effectiveness.

There are several research problems identified system even in the technological sphere in bringing comprehensive solutions. So, in this proposed research aims by solving below problems we can fill above all research problems. First is combining facial landmark detection algorithms with multi-modal convolutional neural networks and ML algorithms in order to accurately predict the attention scores. An accurate detection of facial landmark points, such as the 68 points on a child's face, is also crucial for the precise estimates of gaze and pose. To achieve maximum accuracy, the research also highlights the need for a large dataset of children to train these models effectively.

The review of literature also contains studies that support and hence, demonstrate the effectiveness of the proposed system. For instance, the application of technology, specifically gamification, has resulted in better learning outcomes and consecutive attention in students with learning disabilities [28]. Slow learners who even possess ADHD can be recognized through attention span detection. Refer to [30], on how gamification is used for better learning for ADHD students. Thus, this research mainly focuses on slow learners with the help of a multi-model architecture that combines the evaluation of attention span with interactive game-based learning experiences. This approach meets the requirements of slow learners and provides a very resourceful tool, hence improving the support given to these children at school. By bridging the current gaps in the educational system, especially in developing countries like Sri Lanka, this research will offer a practical, technology-driven tool that enhances educational outcomes for slow learners and supports their overall development.

2. OBJECTIVES

2.1 Main Objective

In Sri Lanka, many students face short attention span challenges which one of the significant reasons for slow learning and some even face conditions like ADHD. These students deserve the same educational opportunities as others, as targeted interventions can increase their learning skills. Therefore, the most important objective here is to facilitate these students by early and timely identification and interventions. Therefore, through a web application this research aims to evaluate the attention span of slow learners using the latest technologies in the domain of face detection, gaze estimation and head posture. Based on this evaluation, it will provide gamified activities in Sinhala to improve attention and cognitive skills. By making the learning process more engaging and adaptive, we hope to improve educational outcomes among slow learners and make their daily lives in school easier. This approach not only supports their academic growth but also fosters their ability to adapt and succeed in life.

2.2 Specific Objectives

- Screening of the attention level and calculating overall attention of child.

Providing a screen test based on an assessment to identify the attention level with multi-model CNN to evaluate the attention span. Here a real time-face identification method is used to recognize and locate the facial landmarks with 68 points to categorize them into five feature groups (nose, mouth, jaw, eye and eyebrows). Regression-based methodology is also used for the model estimation of eye gaze and head pose. Thus, a normalized equation based on [18] to derive attention score for each CNN model.

- Provide real time feedback on attention level.

Providing real time feedback based on the overall score of the screen assessment derived from multi model parameters. Real-time feedback will be provided in an interactive kid-friendly interface that the child gets to know the overall attention for the assessment performed using a motivating approach.

- Generate report with the performed test results.

A report will be generated in the user's profile contributing to the overall attention of the child assessment. This will help the child as well as the teacher to get an overview about the child's attention level.

- Provide gamified activities using Mayer's cognitive theory of multimedia learning.

The goal of this aspect is to provide an intervention to improve the attention level of the child through gamifications. An interactive gamification interface is given to improve focus. The Mayers cognitive theory of multimedia learning will be used in this aspect [8] to make the child more engaging and motivated.

3. SYSTEM METHODOLOGY

The methodology of this research focuses on enhancing learning capabilities by identifying the challenges well in advance, targeted interventions through gamification, and ensuring comfort and accuracy of the screening process through a web-based application. This begins with an in-depth literature review of problems faced by slow learners concerning writing, attention, mathematical skills, and short-term memory. This review is supplemented by a review of current educational technologies and methodologies oriented to the improvement of these exact skills, with a focus on gamified interventions. In so doing, "DiverseMind" offers a more effective and personalized solution by the identification of gaps in existing solutions.

The methodology for the sub-component to identify the attention level and provide them with interventions to enhance their attention span includes a proper plan for requirement analysis, design, and solution development, which would seamlessly fit into the overall platform. High-quality software follows the Software Development Life Cycle (SDLC), but Agile Software Development offers greater flexibility and efficiency. The below Figure 3.1 showcases the agile workflow methodology that has been followed.

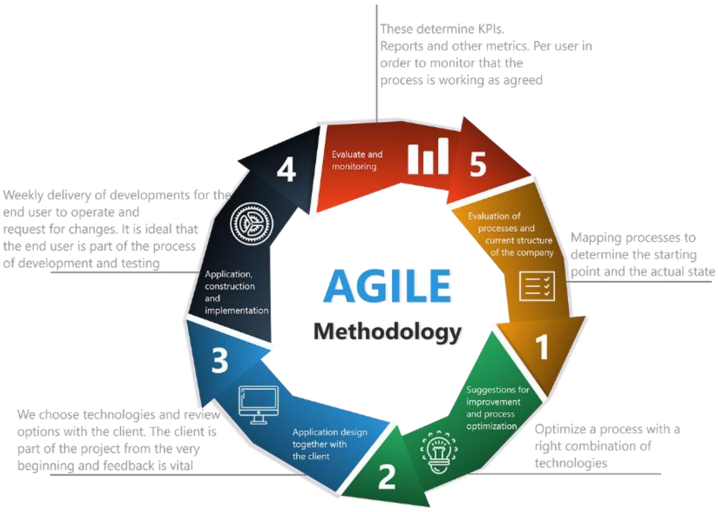


Figure 3.1: Agile workflow

Among Agile methodologies, Scrum is the most widely adopted framework due to its simplicity and ability to enhance collaboration across individuals, teams, and organizations. Its flexibility and adaptability make it the preferred choice over traditional methodologies. “DiverseMind”, followed key Agile practices such as Daily Scrums and Kanban board utilization that played a crucial role. These practices promote teamwork, improve communication, and optimize workflow efficiency, ultimately driving productivity and ensuring the system's successful delivery.

The key pillars of the research methodology, including ML, Image Processing, CNN and deep learning, are utilized for necessary accurate assessment and enhancement of attention spans in slow learners. Analyses of data from multi-sensory inputs, including facial detection, blink count, gaze estimation, and head posture are captured through advanced image processing techniques implemented through a CNN model. Such analyses allow the system to identify patterns that indicate attention deficits. Integration of these cutting-edge technologies can provide very accurate and personalized gamified interventions to improve attention spans among slow learners.

In consequence, "DiverseMind" hopes to enhance the learning capabilities of slow learners tremendously by timely, focused, and engaging interventions while maintaining high standards of accuracy and ethical responsibility.

3.1 System Overview

As illustrated in Figure 3.2, the system overview provides a comprehensive representation of a good overview of the structure of the entire system. The system leverages advanced technologies, including image processing, ML, and multi-model architectures, to analyze students' performance through writing skills, attention span, mathematical skills and working memory and provide tailored interventions.

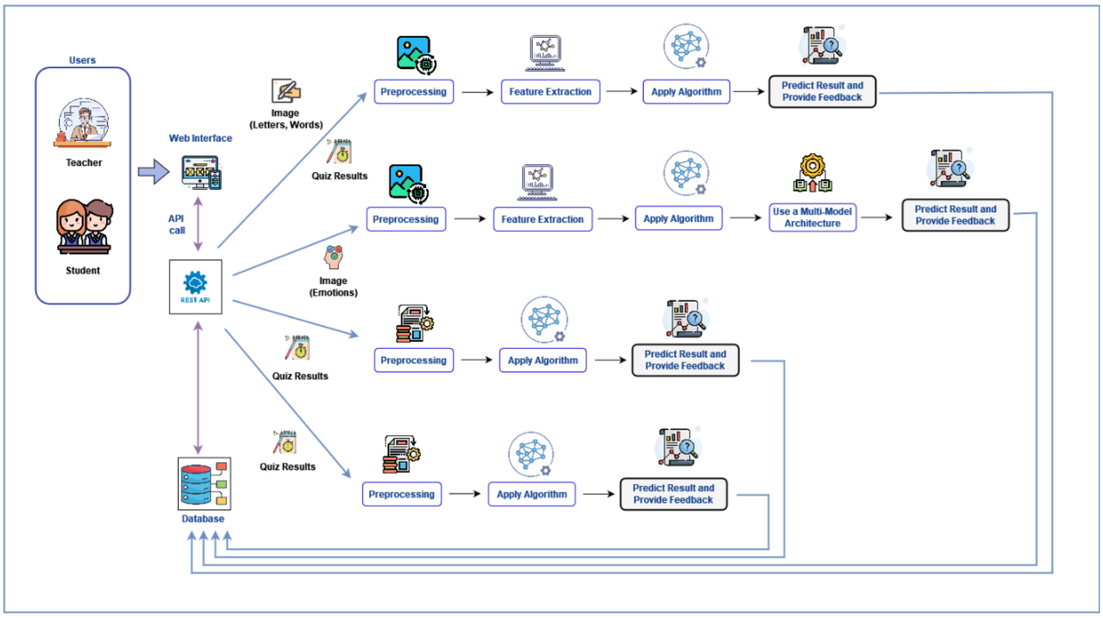


Figure 3.2: System overview diagram

As shown in Figure 3.2, the architecture is based on a web-based platform that incorporates advanced technologies such as FAST APIs (Application Programming Interface), preprocessed data, feature extractions, and algorithms that help support slow learners. In this architecture, a teacher and student interact with a system that has an online interface through which data, results of quizzes or images are processed. The prediction is made by analyzing the claims with different algorithms in an accurate way to retrieve a more personalized feedback portion. This accuracy of results enables targeted interventions for enhancing student learning outcomes.

3.2 Screening and Interventions of Attention Span Evaluation

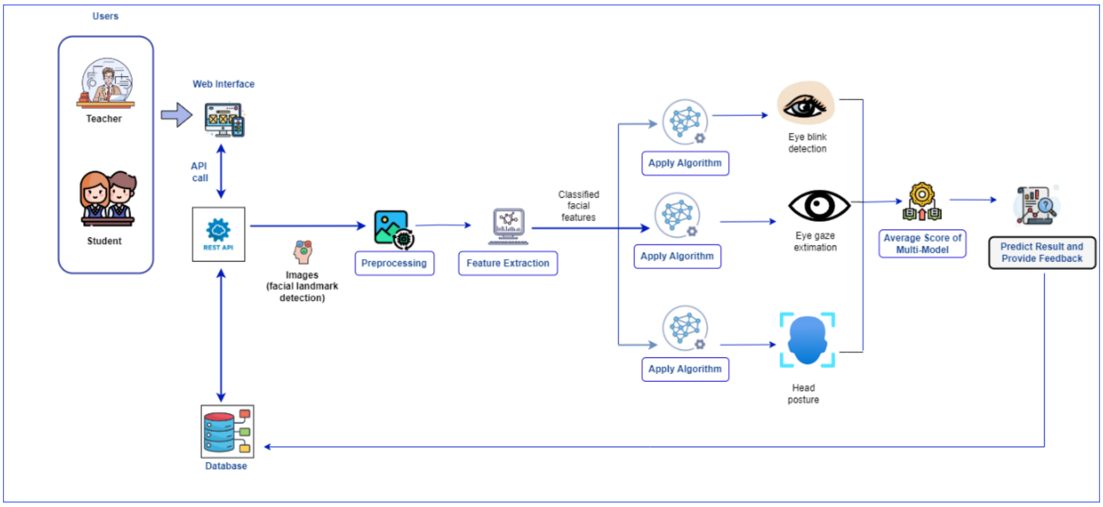


Figure 3.3: Overview of diagram of attention span evaluation

As shown in Figure 3.3, web application assessing the level of attention a child will express, by using the web camera for real-time facial feature extraction. Amongst others, it involves eye blinks, gaze direction and head postures. For these features Media pipe is used with the CNN model for emotion detection captured through the real-time webcam. Thus, the integration of the multi-model predicts the child's attention average score and given a real-time feedback. Through the process, image processing will be utilized together with CNN and Media pipe in order to precisely determine the level of attention of the child. This provides very accurate predictions, achieved by results aggregation through a multi-model approach, thus able to offer personalized feedback and intervention in support of slow learner’s attention levels

The emotion detection is performed using a CNN model trained on grayscale facial images. The goal of this model is to classify emotions into seven distinct categories by extracting spatial features from 48×48 grayscale images while minimizing overfitting and computational complexity.

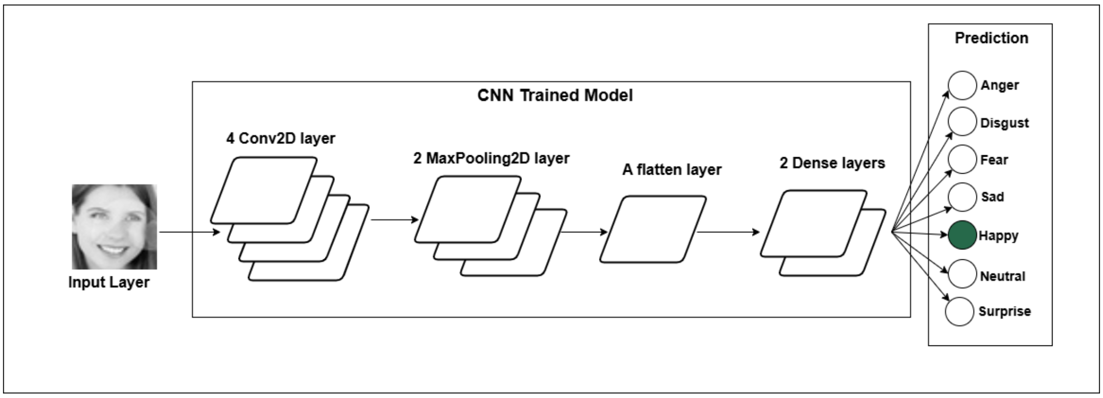


Figure 3.4: CNN architecture of emotion detection

As shown in Figure 3.4, the architecture consists of multiple layers, each serving a specific purpose in feature extraction, down sampling, regularization, and classification. The input layer of the model takes images of size (48, 48, 1), representing grayscale images where each pixel has a single intensity value. This ensures a consistent input format for the network. The core of the model is built using four convolutional layers, each responsible for extracting spatial features from the input images. The first convolutional layer applies 32 filters with a 3×3 kernel, followed by a second convolutional layer with 64 filters and the same kernel size. The third and fourth convolutional layers both use 128 filters with a 3×3 kernel, increasing the depth of the feature maps and allowing the network to capture more complex patterns. Each convolutional layer utilizes ReLU (Rectified Linear Unit) activation, which introduces non-linearity, enabling the network to learn intricate relationships within the image data.

To reduce the spatial dimensions of the feature maps while retaining the most significant information, three max pooling layers are applied after specific convolutional layers. Each max pooling operation uses a 2×2 pool size, which effectively reduces computational complexity by down sampling the feature maps. This pooling mechanism ensures that only the most prominent features are retained, making the model more efficient and robust to variations in facial expressions.

To prevent overfitting and improve the generalization of the model, dropout layers are incorporated at different stages. A dropout rate of 0.25 is applied after the first pooling layer, another 0.25 dropout rate follows the second pooling layer, and a higher dropout rate of 0.5 is introduced after the fully connected dense layer. Dropout works by randomly deactivating a fraction of neurons during training, forcing the network to learn more generalized patterns rather than memorizing specific training examples.

After passing through the convolutional and pooling layers, the feature maps are flattened into a one-dimensional vector using a flatten layer. This transformation allows the extracted features to be fed into fully connected dense layers for classification. The model contains a hidden dense layer with 1024 neurons and ReLU activation, which learns high-level representations of the extracted features. Finally, the output layer consists of 7 neurons with SoftMax activation, which outputs probability distributions for the seven emotion categories, ensuring that the model assigns a confidence score to each class.

The multi- models of facial landmark, eye blinks, gaze direction and head postures integrate through the use of Media pipe and detect the features in real time through the webcam. Facial landmark detection can be performed using Media pipe, Dlib, or MTCNN. Among these, Media pipe is preferred because it provides 468 facial landmarks, offering greater detail compared to Dlib’s 68 points and MTCNN, which lacks comprehensive landmark detection. Eye gaze estimation determines where a child is looking on the screen by measuring the Euclidean distance between the iris and the corners of the eyes. This helps classify the gaze direction as left, right, or center. Blink count detection tracks the number of blinks within a 5-second interval to calculate the average blink rate, which helps assess attention and fatigue levels in real time. According to [34] the average blink rate ranges from 8 to 21 blinks per minute. When deeply focused on a task, the blink rate decreases to approximately 4.5 blinks per minute, whereas low concentration levels can increase it to 32.5 blinks per minute. Head pose estimation analyzes the head’s orientation relative to the screen by calculating angles such as yaw (side-to-side), pitch (up-and-down) and roll. This helps determine whether the user is engaged with the screen or distracted. For accurate detection, the face must be oriented towards the screen to capture facial landmarks, iris position, and blink rate; otherwise, the system proceeds to the next frame.

As shown in Figure 3.5, the attention screening system combines these advanced technologies, facial landmark detection, head position estimation, eye gaze analysis, blink detection, and emotion detection, to comprehensively evaluate a child's focus. Using these metrics, the system calculates an average attention level score and provides a holistic evaluation of the child's attentiveness.

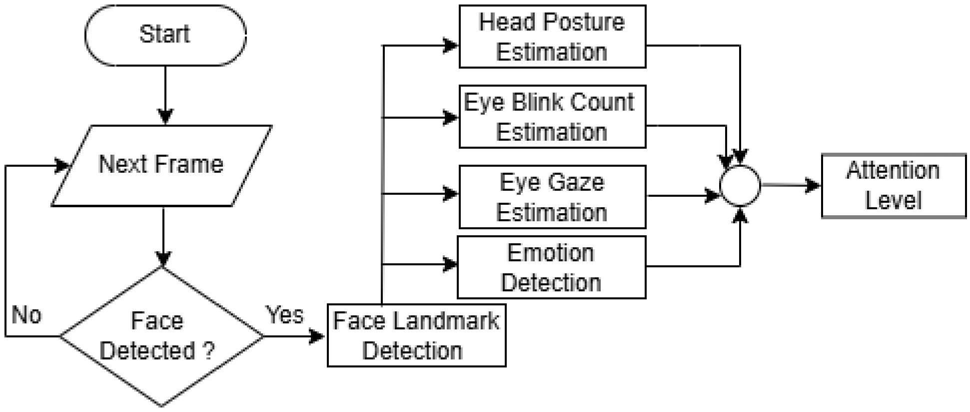


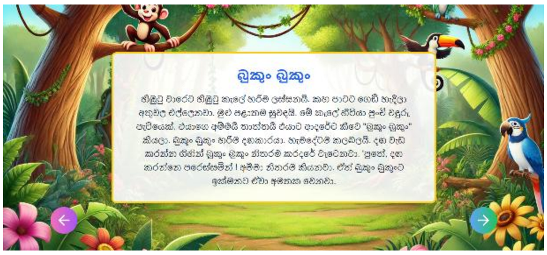
Figure 3.5: High level diagram for attention level assessment

During the screening process, the child undergoes a structured reading assessment aimed at evaluating their attention level while interacting with the screen. In this assessment, the child is presented with a passage displayed on the screen as shown in Figure 3.6.a and Figure 3.6.b. The task requires them to focus, read through the passage carefully, and comprehend its content.

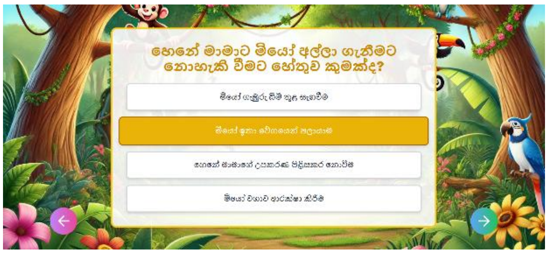
To measure their attention and engagement, the system monitors the time spent reading and how consistently they follow through with the passage without significant pauses or distractions. After reading, the child is required to answer a series of Multiple-Choice Questions (MCQs) based on the passage as shown in Figure 3.6.c and Figure

3.6.d. These questions assess their understanding of the content while also indirectly evaluating their ability to maintain focus.

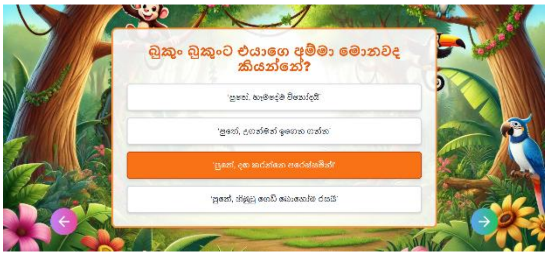
(a) Reading passage 1



(b) Reading passage 2



(c) Screening quiz for passage 1

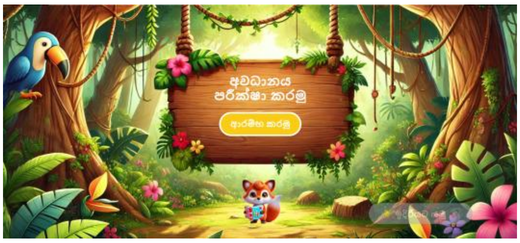


(d) Screening quiz for passage 2

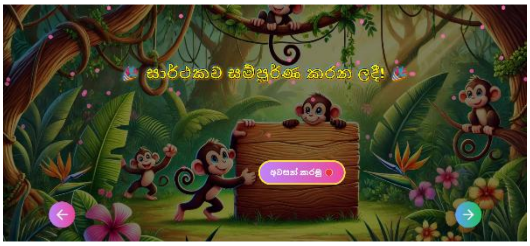
Figure 3.6: Attention screening

By analyzing the reading behavior, the system can determine the child's attention level. This assessment provides valuable insights into their reading skills, comprehension abilities, and overall attentiveness, which can further guide personalized interventions to support their learning needs.

As shown in Figure 3.7.a to enable real-time attention tracking, the child is required to turn on the webcam before starting the assessment. Once the assessment is complete, clicking the camera off button stops the monitoring process as shown in Figure 3.7.b and ends the attention tracking.



(a) Camera on interface

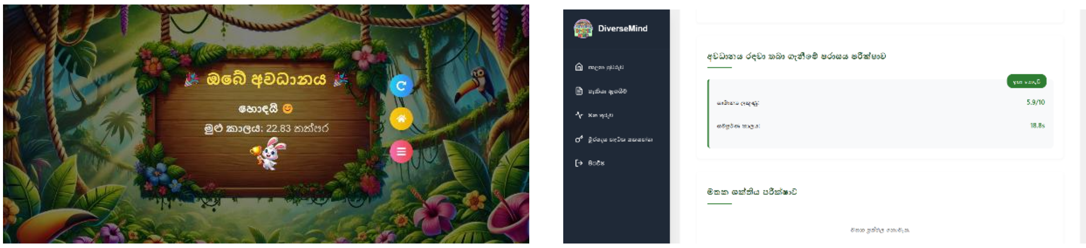


(b) Camera off interface

Figure 3.7 : Attention screening camera on and off interface

When the webcam is activated, a FAST API request is sent to the backend server using FastAPI, initiating the attention monitoring system. Similarly, when the child clicks the camera off button, another API request is sent to disable tracking. Throughout the assessment, key attention metrics are continuously analyzed in real time. These metrics are automatically stored in the MongoDB database when the assessment concludes, and the webcam is turned off.

For each assessment attempt, the child receives instant feedback based on their attention score as shown Figure 3.8.a. This allows for real-time insights into their focus levels, helping them improve their engagement during reading tasks. Additionally, the total time spent on the assessment is calculated and displayed to the child.



(a) Attention results feedback (b) Teacher dasboard attention results report

Figure 3.8 : Attention feedback and report

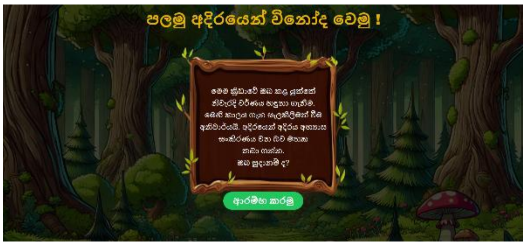
A comprehensive report, accessible only to teachers, provides insights such as the child's attention focus status, total assessment duration, and average attention score as shown in Figure 3.8.b. This report enables educators to assess the child's reading engagement, track attention patterns, and implement effective interventions when necessary.

To enhance children's attention levels, a series of gamified intervention activities are provided as shown in Figure 3.9, each designed to target specific aspects of attention. The intervention consists of three distinct games, with each game structured into three progressively challenging levels. Starting from level one, the difficulty increases as children advance (see Figure 3.10.a, Figure 3.11.a and Figure 3.12.a) requiring greater concentration and cognitive effort.

Figure 3.9: Attention intervention Menu

As shown in Figure 3.10, “Choose the Color” gamification is designed to improve focused attention and cognitive control. Before starting, the child is presented with an instruction page (see Figure 3.10.b) that explains the game mechanics, ensuring they understand the objective. The game is time-based and tracks both the total score and the best score to encourage progress as shown in Figure 3.10.c. During the game, the child must quickly recognize and select the correct color displayed on the screen, reinforcing their ability to stay focused and make rapid decisions. At the end, the game concludes with an interactive interface featuring engaging animations (see Figure 3.10.d), making the experience enjoyable and rewarding.

(a) Level page



(b) Instruction page

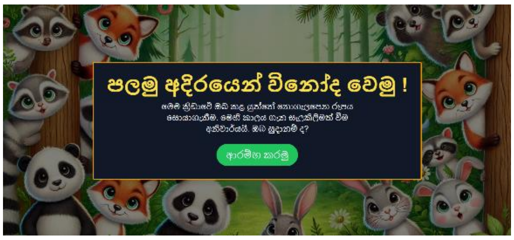
(c) Choose the color game

(d) Game Score

Figure 3.10: Choose the color intervention game

As shown in Figure 3.11, “Spot the Differences” gamification focuses on enhancing selective attention by training children to observe and identify small differences between seemingly identical images as shown in Figure 3.11.c. As shown in Figure 3.11.b, the game begins with an instruction page, guiding the child on how to play effectively. Each round is limited to 15 seconds, requiring the child to analyze the images quickly while filtering out distractions. Once the game ends, the child receives motivational feedback to encourage continued effort and engagement, making the experience both challenging and rewarding as shown in Figure 3.11.d.

a) Level page



(b) Instruction page

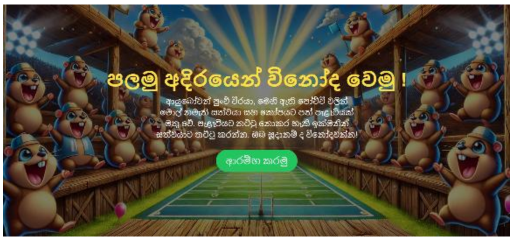
(c) Spot the difference game

(d) Game Score

Figure 3.11: Spot the difference intervention game

As shown in Figure 3.12, “Whack-a-Mole” gamification is designed to strengthen sustained and divided attention along with impulse control. The game starts with an instruction page (see Figure 3.12.b) to ensure the child understands the rules before playing. As the game begins, moles appear randomly on the screen, and the child must react quickly to hit them while avoiding mistakes as shown in Figure 3.12.c. If the child accidentally clicks on a plant instead of a mole, the game ends immediately, displaying both the total score and the best score as shown in Figure 3.12.d. To create an immersive and engaging experience, background music is played throughout the game, keeping the child entertained and motivated.

(a) Level page



(b) Instruction page

(c) Whack the mole game

(d) Game Score

Figure 3.12: Whack the mole intervention game

These three interactive games are designed with user-friendly interfaces that make them highly interactive and engaging. By integrating cognitive training into a fun and gamified format, these activities help children develop essential attention skills in an enjoyable and stimulating way.

3.3 Requirements

3.3.1 Functional requirement

- Registration and Login - The user must be able to register and log in securely and be authenticated and authorized into the system.
- Initiate the attention span evaluation assessment - The valid user who is authorized and authenticated into the system can attempt the assessment.

- Real-time attention span monitoring - During the evaluation the system analyses the student's attention through various parameters such as facial detection, blink rate, gaze direction, and head posture.
- View the attention span results - After the evaluation is completed the user can view the results through reports.
- Real-time feedback generation - According to the attention span results feedback are generated.
- Gamification - Gamified activities are provided for intervention.
- UI based in Sinhala language - The application is supported in Sinhala language for the user.

3.3.2 Use cases diagram

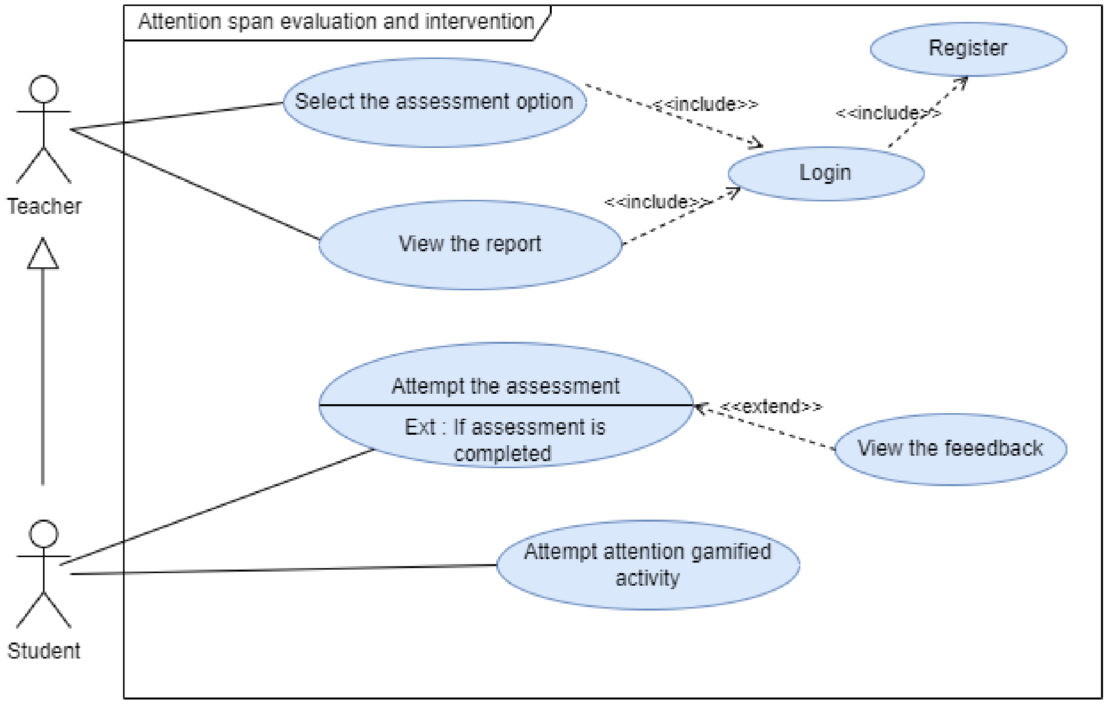


Figure 3.13: Attention span evaluation and intervention user case diagram

3.3.3 System use cases scenarios

Table 3.1: Attempt the assessment user case scenario

Use case ID	001
Use case name	Attempt the assessment
Primary actor	Student
Pre-condition	Users should have valid login credentials
Main flow	1. User login. 2. Select the screening assessment. (ඇගයීම) option from the main menu. 3. Select the attention span assessment option (“අවධානය පරීක්ෂා කරමු) from screening menu. 4. Click on the start button (ආරම්භ කරමු) and the web camera gets activated. 5. Proceed with the assessment by clicking on the right-side arrow button after completion. 6. Click on the end button (අවසන් කරමු) option once completed with the assessment and the web camera gets deactivated. 7. The real-time feedback is displayed on the screen (ඉතා හොදයි!/හොදයි!/උනන්දු විය යුතුයි!).

Extension	5a. If the right arrow button is clicked without completing the task, it remains disabled, preventing further progress in the assessment.

Table 3.2: View the report user case scenario

Use case ID	002
Use case name	View the Report
Primary actor	Teacher
Pre-condition	The user should have valid conditions. The student should have attempted the assessment
Main flow	1. Users need to log in. 2. Navigate to the teachers' dashboard. 3. Move to the enrolled students' option in the dashboard. 4. Click on the report option on the right side of the student's name. 5. View the report details.

Table 3.3: Attempt the gamified attention activity user case scenario

Use case ID	003
Use case name	Attempt the gamified attention activity
Primary actor	Student
Pre-condition	The students identified as short attention span.
Main flow	1. Users need to log in.

	2. Click on intervention activities (“මඟ හුරුව” from the main menu. 3. Click on the attention span intervention activity (අවධානය සඳහා මගහුරුව) from the intervention menu. 4. The user can attempt any option of the provided three gamified activities that have three levels of each (වර්ණ හඳුනා ගනිමු ,නොගැලපෙන රූපය සොයමු ,ඉලක්කය හරිද බලමු). 5. Once ready to attempt the game click on the start button (ආරම්භ කරමු). 6. Game results displayed on screen when completed all rounds.

3.3.4 Non-functional requirement

- Reliability - Attention span evaluations should be provided consistently and accurately. It needs to be reliable enough not to fail or stop working in between processes, especially in crucial applications like real-time monitoring and result generation.
- Scalability - The system should be able to handle increasing amount of users. As the users grow with time it should be able to scale and accommodate additional amount of users.

- Performance- The attention span data needs to be processed and analyzed in real-time with very minimal latency. Facial detection, the counting of blinks, gaze estimation, and head posture analysis need to occur seamlessly without allowing users to experience delays.
- Security - It is important that the system ensures all user data, most especially sensitive student information and evaluation results, stored and transmitted securely. In this respect, strong mechanisms of identification and authentication have been enabled to protect unauthorized access.
- Availability - The system should be available for use 24/7, with minimal downtime.
- Maintainability - The system should be maintainable and easy for updates, bug fixes and enhancements.
- Usability -The system should be user-friendly and easy for users to navigate.
- Accuracy - The system shall ensure high accuracy in attention span evaluation by the adoption of reliable algorithms and precise methods of data analysis.

3.3.5 System requirement

- **Software Requirements**
  - User-end
    - ➢ Web browser (E.g. Google Chrome, Firefox, Safari)
  - Developer end
    - ➢ **IDEs -** Microsoft Visual Studio
    - ➢ **Programming Language -** Python 3.8 or above, ReactJS 18
    - ➢ **Framework** – FAST API for creating APIs
    - ➢ **Libraries** - TensorFlow 2.13 or above
    - ➢ **Database -** MongoDB
    - ➢ **Version Control -** Git and GitHub

- **Hardware Requirements**
  - Web camera

3.3.6 Challenges

Throughout the research, we encountered several challenges that required careful problem-solving and adaptation. Below are some of the key challenges that were faced and resolved successfully.

- Obtaining permission for data collection - Securing approval from schools for data collection proved to be a significant challenge due to the strict ethical and administrative processes involved. It was necessary to understand and adhere to the correct protocols for data collection in an educational setting. Permission had to be formally obtained from the Zonal Education Office of Sri Jayawardhanapura, which required thorough documentation and justifications regarding the purpose, ethical considerations, and benefits of the study. This process involved multiple discussions and approvals before gaining authorization.
- Finding a suitable medical expert - An essential aspect of the project was securing guidance from a qualified doctor to ensure the process was carried out with proper medical and psychological considerations. However, identifying an appropriate professional who could commit to providing the necessary insights was a challenging task. After extensive search and consultations with various experts, a suitable doctor was finally found, allowing the team to proceed with a well-informed approach.
- Selecting the appropriate dataset - Choosing an appropriate dataset for training the emotion detection model was a critical step, as the quality and relevance of the data directly impacted the model's accuracy and effectiveness. The team explored multiple datasets to assess their compatibility with the project's objectives. Several datasets were evaluated based on their structure, sample diversity, and applicability to real-world scenarios before finalizing the most suitable one for model training.
- Technical and implementation challenges - The project involved working within the sphere of data science and machine learning, which was relatively

new to the team. Understanding the concepts of training models, preprocessing data, and optimizing performance required learning from scratch. The team had to conduct extensive research, experiment with different methodologies, and familiarize themselves with various technological tools and frameworks to ensure effective implementation.

- Time management and coordination- Managing time efficiently was one of the most demanding aspects of the project, as multiple tasks such as implementation, presentations, and documentation which were needed to be handled simultaneously within strict deadlines. To streamline workflow and enhance productivity, the team adopted Microsoft Planner, a project management tool, to track tasks, deadlines, and daily scrums. Coordinating with all team members and ensuring alignment with each individual's availability was crucial to maintaining steady progress.
- Challenges during the testing phase - Minor conflicts arose during the testing phase, particularly in selecting a suitable school and obtaining their permission to conduct testing. Schools had specific time constraints and academic schedules, which had to be considered when planning the testing sessions. Aligning the testing schedule with the school's availability required flexibility, effective communication, and negotiation to ensure minimal disruption to their daily operations.

Each of these challenges required strategic problem-solving, adaptability, and continuous learning to ensure the successful completion of the project.

3.4 Commercialization Aspects of the Products

“DiverseMind” is an educational platform designed to support slow learners who experience difficulties in writing, attention, mathematical skills, and working memory. Unlike existing tools that address only one aspect of learning difficulties, “DiverseMind” provides an integrated approach by combining multiple assessments and personalized interventions in a single system. Currently, there is no similar comprehensive tool specifically designed for slow learners available in Sri Lanka, making this a unique and much-needed solution.

The effectiveness and relevance of “DiverseMind” have been evaluated and endorsed by experienced educators and professionals. Experts such as Mr. Gamini Udhaya Kumara (Primary School In-Service Advisor, Zonal Office Kurunegala), Mrs. Nadee Gunarathne (Vice Principal, Sri Subhuthi National School Battaramulla), and Mrs. C.A.D. Dhaya Nishanthi (Primary School In-Service Advisor, Sri Jayawardenepura Education Zone) have confirmed that the assessments and interventions included in the platform are appropriate for supporting slow learners. Their recommendations strengthen the credibility of “DiverseMind” and its alignment with the current educational framework in Sri Lanka.

3.4.1 Target market and pricing model

“DiverseMind” follows a subscription-based pricing model to ensure affordability and accessibility for different user groups. The pricing plans are structured as shown in Table 3.4.

Table 3.4: DiverseMind subscription plans and pricing

Plan	Feature	Pricing
Individual Plan	Suitable for single users, providing access to all features, updates, and technical support.	Rs. 1000 per month

Family Plan	Supports up to three students, offering multi student management with feature updates and technical support.	Rs. 2500 per month
Classroom Plan	Designed for schools, allowing up to 40 students, best suited for any school or educational institution. It consists of multi-student management, feature updates, and ongoing technical support.	Rs. 5000 per month
Support for Underprivileged Families	Special discounts will be provided to ensure accessibility for economically disadvantaged families.	Special Pricing

“DiverseMind” will be introduced to the market through a comprehensive digital marketing strategy using Facebook, Instagram, and YouTube. These platforms will be used to create awareness through educational content, success stories, and interactive demonstrations of the system’s features.

To encourage adoption, collaborations with schools, government bodies, and educational organizations will be established. Free assessments will be conducted to demonstrate the effectiveness of the platform and build trust among parents and teachers. Additionally, expert endorsements from educators and specialists will help position “DiverseMind” as a validated and reliable educational tool.

Furthermore, the platform will be introduced in teacher training programs and educational workshops to increase awareness and encourage adoption within classrooms. These efforts will ensure that “DiverseMind” reaches both individual users and educational institutions effectively.

3.5 Testing and Implementation

3.5.1 Implementation

“DiverseMind” is a software solution designed as a web application. The frontend development of the application was carried out using React Native and JavaScript, enabling a responsive and interactive user interface. A MongoDB cluster serves as the database, providing efficient data storage and retrieval capabilities for handling user information, activity logs, and learning progress. The backend was developed using Python, ensuring smooth communication between the frontend and database while handling complex processing tasks. FAST API is utilized to establish seamless communication between the backend and frontend. The machine learning models were trained using Jupyter Notebook, a versatile environment that facilitated interactive coding, visualization, and iterative model development. OpenCV, a powerful computer vision library, was utilized for real-time detection using the device's webcam. This enabled the application to monitor users’ attention spans effectively by tracking facial features, gaze direction, and head movements. Extensive image processing techniques were applied to enhance the quality of input data and optimize the accuracy of detection models. These techniques ensured that the system could reliably interpret visual information and deliver insightful feedback to users. TensorFlow, a popular deep learning framework, was used alongside OpenCV for model training and deployment, allowing for accurate and efficient attention span detection.

The implementation of the screening of the attention was an integration of several models. For the emotion detection the concept deep learning concept was utilized and CNN chosen as the algorithm. The emotion detection dataset consisted of 28709 training images and 7178 testing images as shown in Figure 3.14.

```
Found 28709 images belonging to 7 classes.
Found 7178 images belonging to 7 classes.
```

Figure 3.14: Total number of images of the dataset

Data preprocessing is a crucial step in preparing images for training a deep learning model, as it ensures consistency and optimizes performance. As shown in Figure 3.15, the ImageDataGenerator class is used to handle preprocessing and augmentation tasks efficiently. By setting the rescale parameter to 1./255, pixel values are normalized from a range of [0, 255] to [0, 1].

```
[2]: # Initialize image data generator with rescaling
     train_data_gen = ImageDataGenerator(rescale=1./255)
     validation_data_gen = ImageDataGenerator(rescale=1./255)
```

Figure 3.15: ImageDataGenerator function

This scaling transformation ensures that the model receives inputs with smaller and consistent values, which stabilizes and accelerates the training process by preventing issues arising from large gradient values. This step is essential for efficient convergence during model training. Separate ImageDataGenerator instances are created for the training and validation datasets, ensuring both datasets undergo consistent preprocessing.

```
[3]: # Preprocess all train images
     train_generator = train_data_gen.flow_from_directory(
             'data/train',          # Path to training dataset
             target_size=(48, 48), # Resize all images to 48x48
             batch_size=64,         # Batch size
             color_mode="grayscale", # Use grayscale images
             class_mode='categorical') # Categorical labels

     # Preprocess all validation images
     validation_generator = validation_data_gen.flow_from_directory(
             'data/test',           # Path to validation dataset
             target_size=(48, 48), # Resize all images to 48x48
             batch_size=64,         # Batch size
             color_mode="grayscale", # Use grayscale images
             class_mode='categorical') # Categorical labels

     Found 28709 images belonging to 7 classes.
     Found 7178 images belonging to 7 classes.
```

Figure 3.16: Data preprocessing code snippet

After data preprocessing, the flow_from_directory method from ImageDataGenerator is employed to load images directly from directory structures where subfolders represent each class as shown in Figure 3.16. Using the target_size parameter, images are resized to 48x48 pixels, reducing computation time while retaining essential facial features. Setting the batch_size to 64 ensures the model processes 64 images in a single step, optimizing memory usage and training speed. The color_mode is set to "grayscale" to convert images from RGB to grayscale as shown in Figure 3.17, reducing the complexity of input data while preserving important details. This simplification is particularly effective for facial recognition, as color information is less relevant compared to intensity variations.

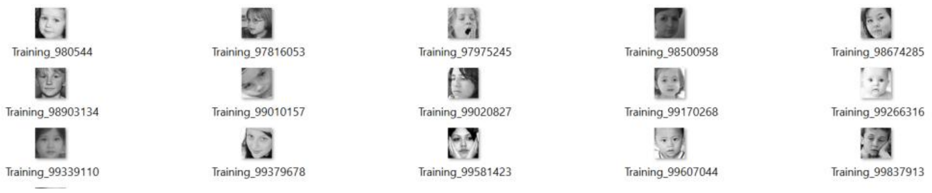


Figure 3.17: Grayscale images data

The CNN model is designed for emotion classification from grayscale images of size 48x48 pixels. It starts with an input layer and progresses through several convolutional layers (32, 64, and 128 filters) to extract increasingly complex features. ReLU activation is used for non-linearity, and MaxPooling2D layers reduce spatial dimensions. Dropout layers (0.25 and 0.5) are included to prevent overfitting by randomly deactivating neurons during training.

After convolutional layers, the Flatten layer prepares the data for fully connected dense layers, with the first dense layer having 1024 neurons and ReLU activation. A final output layer with seven neurons corresponds to seven emotion classes as Angry, Disgust, Fear, Happy, Sad, Surprise, and Neutral, using the softmax activation function for multi-class classification as shown in Figure 3.18.

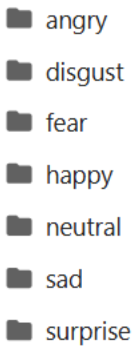


Figure 3.18: Categorization of emotions

The model is trained with one-hot encoding and minimizes error through categorical cross-entropy, enabling it to classify emotions based on the extracted features.

The python code is a real-time head pose estimation system that uses a webcam and the Media pipe library to track a user's face and determine if they are paying attention or distracted. It works by continuously capturing video frames from the webcam and processing each frame to detect facial landmarks. The face mesh model from Media pipe identifies specific landmark points on the face, particularly around the eyes, nose, and cheeks, which are then used to estimate the orientation of the head in 3D space. By calculating the yaw (left-right rotation), pitch (up-down rotation), and roll (tilting sideways) as shown in Figure 3.19, the system can tell the direction in which the person is looking. Therefore, if the head is not oriented towards the screen the next frame is requested.

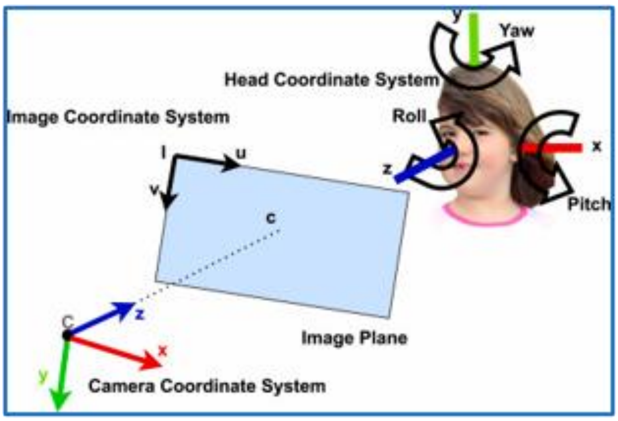


Figure 3.19: Head pose estimation relative to screen

The eye gaze tracking tracks a person's eye movement using a webcam and Media pipe's Face Mesh model. It captures real-time video, detects facial landmarks specifically the right eye and iris and calculates the iris position relative to the eye corners. Based on this, it determines if the person is looking left, right, or center.

The final integration model of eye blink is designed to detect and count eye blinks in real-time using a webcam. It leverages Media pipe's Face Mesh model to identify facial landmarks, particularly focusing on the eyes. When the webcam captures each frame, the frame is resized and converted to RGB because Media pipe requires RGB format for processing. The model then detects landmarks on the face, and specific landmark indices around the left and right eyes are used to determine whether a blink has occurred.

To detect a blink, the code calculates the Euclidean distance between points on the eye as shown in Figure 3.20. The horizontal and vertical distances are used to compute a blink ratio for each eye. If the average of these ratios exceeds a threshold, it indicates that the eyes are closed. If this condition is sustained for a few frames, it's considered a blink.

```
def blink_ratio(landmarks, right_eye_indices, left_eye_indices):
    """
    Calculates the blink ratio using landmarks for both eyes.
    """
    # Right eye
    rh_distance = euclidean_distance(landmarks[right_eye_indices[0]], landmarks[right_eye_indices[8]])
    rv_distance = euclidean_distance(landmarks[right_eye_indices[12]], landmarks[right_eye_indices[4]])
    # Left eye
    lh_distance = euclidean_distance(landmarks[left_eye_indices[0]], landmarks[left_eye_indices[8]])
    lv_distance = euclidean_distance(landmarks[left_eye_indices[12]], landmarks[left_eye_indices[4]])
    # Calculate ratios
    right_ratio = rh_distance / rv_distance
    left_ratio = lh_distance / lv_distance
    return (right_ratio + left_ratio) / 2
```

Figure 3.20: Euclidean distance calculation code snippet

Thus, the integration of the models, the average attention metric is calculated to identify the child’s attention level. Thus, the average attention metric predicts the attention level into 3 categories of focused, moderately focused and Not Focused.

3.5.2 Testing

To verify and validate the functionality of the web application, thorough testing is essential. Initially, test cases were designed to cover all features of the application. To ensure accuracy, both functional and non-functional testing were performed. Below are some examples of test cases conducted for the attention span component.

Table 3.5: Predict the attention level (normal child) test case

Test case ID	TC01
Test Case Scenario	Predict the attention level - User has good attention levels (normal child).
Test input data	Face landmark, facial expressions, blink count, gaze direction and head posture detected through real-time web camera.
Test procedure	1. Select the screening assessment (ඇගයීම) option from the main menu. 2. Select the attention span assessment option (“අවධානය පරීක්ෂා කරමු) from screening menu. 3. Click on the start button (ආරම්භ කරමු) and the web camera gets activated. 4. Proceed with the assessment by clicking on the right-side arrow after completion. 5. Click on the end button (අවසන් කරමු) option once completed with

	the assessment and web camera gets deactivated. 6. The real-time feedback is displayed on the screen (ඉතා හොඳයි!/හොඳයි!/උනන්දු විය යුතුයි!).
Expected output	The user should receive the feedback as “ ඉතා හොඳයි!”.
Actual output	The user received the feedback as “ ඉතා හොඳයි!”.
Test result	Pass

Table 3.6: Predict the attention level test case (short attention span)

Test case ID	TC02
Test Case Scenario	Predict the attention level - User does not have good attention levels (a slow learner with attention span difficulties).
Test input data	Face landmark, facial expressions, blink count, gaze direction and head posture detected through real-time web camera.
Test procedure	1. Select the screening assessment (ඇගයීම) option from the main menu. 2. Select the attention span assessment option (“අවධානය පරීක්ෂා කරමු) from screening menu. 3. Click on the start button (ආරම්භ කරමු) and the web camera gets activated. 4. Proceed with the assessment by clicking on the right-side arrow after completion.

	5. Click on the end button (අවසන් කරමු) option once completed with the assessment and web camera gets deactivated. 6. The real-time feedback is displayed on the screen (ඉතා හොඳයි! /හොඳයි! /උනන්දු විය යුතුයි!).
Expected output	The user should receive the feedback as “උනන්දු විය යුතුයි!”.
Actual output	The user received the feedback as “උනන්දු විය යුතුයි!”.
Test result	Pass

Table 3.7: Attempt “choose the color” intervention test case (success)

Test case ID	TC03
Test Case Scenario	Attempt the “choose the color” (වර්ණ හඳුනා ගනිමු) game for attention intervention - User has successfully completed the game within the given time with highest score.
Test input data	Results from the game.
Test procedure	1. Select the intervention (මහ හුරුව) button in the main menu. 2. Select the writing interventions (අවධානය සඳහා මගහුරුව) option in the intervention menu. 3. Select the “choose the color” (වර්ණ හඳුනා ගනිමු) game from the attention intervention menu.

	4. Click on the start (ආරම්භ කරමු) and proceed with the attention intervention game. 5. Proceed with the game and choose the correct color within the given time. 6. Directs to the scoreboard which displays the score, rewards earned and the best score.
Expected output	The user should receive a congratulation message when the user gets a 100% score.
Actual output	The user received the congratulation message.
Test result	Pass

Table 3.8: Attempt “choose the color” intervention test case (not success)

Test case ID	TC04
Test Case Scenario	Attempt the “choose the color” (වර්ණ හඳුනා ගනිමු) game for attention intervention - User has failed to successfully complete the all five questions within the given time with highest score.
Test input data	Results from the game.

Test procedure	1. Select the intervention (මඟ හුරුව) button in the main menu. 2. Select the writing interventions (අවධානය සඳහා මගහුරුව) option in the intervention menu. 3. Select the "choose the color" (වර්ණ හඳුනා ගනිමු) game from the attention intervention menu. 4. Click on the start (ආරම්භ කරමු) and proceed with the attention intervention game. 5. Proceed with the game and choose the correct color within the given time. 6. Directs to the scoreboard which displays the score, rewards earned and the best score.
Expected output	The user should receive the retry message
Actual output	The user received the retry message.
Test result	Pass

Table 3.9: Attempt "spot the difference" intervention test case (success)

Test case ID	TC05
Test Case Scenario	Attempt the "spot the difference" (නොගැලපෙන රූපය සොයමු) game for attention intervention -User has successfully completed the game within the given time with highest score.
Test input data	Results from the game.

Test procedure	1. Select the intervention (මඟ හුරුව) button in the main menu. 2. Select the writing interventions (අවධානය සඳහා මගහුරුව) option in the intervention menu. 3. Select “spot the difference” (නොගැලපෙන රූපය සොයමු) game from the attention intervention menu. 4. Click on the start (ආරම්භ කරමු) and proceed with the attention intervention game. 5. Proceed with the game and choose the different image within the given time. 6. Directs to the scoreboard which displays the score, feedback and earned stars.
Expected output	The user received the feedback as “ ඉතා හොඳයි!” with 5 stars.
Actual output	The user received “ ඉතා හොඳයි!” with 5 stars.
Test result	Pass

Table 3.10: Attempt “spot the difference” intervention test case (not success)

Test case ID	TC06
Test Case Scenario	Attempt the “spot the difference” (නොගැලපෙන රූපය සොයමු) game for attention intervention - User has failed to successfully complete the all five questions within the given time with highest score.
Test input data	Results from the game.

Test procedure	1. Select the intervention (මඟ හුරුව) button in the main menu. 2. Select the writing interventions (අවධානය සඳහා මගහුරුව) option in the intervention menu. 3. Select “spot the difference” (නොගැළ පෙන රූපය සොයමු) game from the attention intervention menu. 4. Click on the start (ආරම්භ කරමු) and proceed with the attention intervention game. 5. Proceed with the game and choose the different image within the given time. 6. Directs to the scoreboard which displays the score, feedback and earned stars.
Expected output	The user received the feedback as “ උනන්දු විය යුතුයි!” with no stars earned.
Actual output	The user received the feedback as “ උනන්දු විය යුතුයි!” with no stars earned.
Test result	Pass

Table 3.11: Attempt “whack the mole game” interventions test case (success)

Test case ID	TC07
Test Case Scenario	Attempt the “whack the mole game"(ඉලක්කය හරිද බලමු) game for attention interventions. -User clicks on the targeted mole and scores 100 or more marks to earn 3 stars.
Test input data	Results of the game

Test procedure	1. Select the intervention (මඟ හුරුව) button in the main menu. 2. Select the writing interventions (අවධානය සඳහා මගහුරුව) option in the intervention menu. 3. Select the “whack the mole game"(ඉලක්කය හරිද බලමු) game for attention interventions 4. Click on the start (ආරම්භ කරමු) and proceed with the attention intervention game. 5. The user needs to click on the targeted mole correctly to earn marks but if the plant gets clicked by mistake the game gets over. 6. The scoreboard displays the earned stars, score and best score.
Expected output	The user clicks on the targeted mole and scores 100 or more marks to earn 3 stars and also the score and best score is displayed.
Actual output	The user received 100 or more marks with the best score and earned 3 stars.
Test result	Pass

Table 3.12: Attempt “whack the mole game" interventions test case (not success)

Test case ID	TC08
Test Case Scenario	Attempt the “whack the mole game"(ඉලක්කය හරිද බලමු) game for attention interventions – The user clicks on the plant and the game gets over.

Test input data	Results of the game.
Test procedure	1. Select the intervention (මඟ හුරුව) button in the main menu. 2. Select the writing interventions (අවධානය සඳහා මගහුරුව) option in the intervention menu. 3. Select the “whack the mole game"(ඉලක්කය හරිද බලමු) game for attention interventions 4. Click on the start (ආරම්භ කරමු) and proceed with the attention intervention game. 5. The user needs to click on the targeted mole correctly to earn marks but if the plant gets clicked by mistake, the game gets over. 6. The scoreboard displays the earned stars, score and best score.
Expected output	The users game gets over when the plant is clicked by accident.
Actual output	The user clicks on the plant by accident and the game gets over.
Test result	Pass

3.5.3 Functional testing

- Unit testing - To ensure that all individual components function correctly, unit testing is conducted separately for the attention screening process and gamified activities for attention interventions, real-time feedback and report generation.

- Integration testing - Individual components of the screening process, interventions, real-time feedback and report generation are integrated and tested as a group to ensure seamless functionality and data flow.

- System testing - The entire system, including the screening and intervention components, is tested as a whole to verify that results align with expected performance norms. This assessment includes the modules responsible for evaluating writing skills and providing interventions, assessing attention span and providing corresponding interventions, analyzing mathematical abilities and providing targeted support, and evaluating working memory and providing interventions.

- Regression testing - Conducted to confirm that existing features remain functional after implementing modifications or bug fixes, ensuring system stability and reliability over time.

3.5.4 Non-functional testing

- Usability testing - Usability testing is conducted to ensure that the platform effectively supports students with varying attention spans. Special focus is placed on designing an engaging and interactive experience that helps maintain students' focus. Elements such as the fonts, buttons, and overall layout are carefully structured to be visually appealing, easy to navigate, and free from distractions. The interface is optimized for children aged 9 years, incorporating clear instructions, intuitive controls, and appropriate visuals to sustain their attention throughout the activities.

- Security testing - Security testing ensures that all data related to students' attention span assessments, including activity responses and test results, are securely stored and processed. The system complies with data protection guidelines to safeguard sensitive information, preventing unauthorized access or data breaches. Encryption and secure authentication mechanisms are implemented to protect student records while maintaining seamless access for teachers and authorized personnel.
- Performance Testing - Performance testing ensures that the platform runs smoothly without delays or lag, maintaining student engagement during attention span activities. The system is evaluated for fast response times, minimal loading interruptions, and the ability to handle multiple users simultaneously. Optimized animations, quick feedback mechanisms, and seamless transitions help sustain students' focus, preventing distractions caused by technical slowdowns.
- Compatibility testing - Compatibility testing ensures that the attention span assessment functions seamlessly across multiple browsers and devices. The system is tested to maintain consistent performance, responsive design, and intuitive interactions, allowing students to engage with activities without technical interruptions. This ensures that the platform remains accessible and effective for children aged 9, regardless of the device or browser they use.

3.6 Work Breakdown Structure

The workload of the research project is shown in Appendix C.

3.7 Gantt Chart

The Gantt chart of the research project is shown in Appendix B.

4. RESULT AND DISCUSSION

4.1 Results

The attention span evaluation system was designed to measure user engagement based on multiple parameters, including facial landmark, head pose estimation, eye gaze tracking, blink rate detection, and emotion recognition.

The integration of Media pipe enabled highly precise facial landmark detection, significantly improving the accuracy of attention level analysis compared to traditional methods like Dlib. The system successfully detected 468 landmark points with 3D coordinates, offering a significant improvement in accuracy compared to Dlib, which only detects 68 2D points. The 3D landmark detection provided a more precise facial mapping, allowing for better analysis of facial expressions, movements, and micro-expressions. The model showed robust performance even in challenging conditions, such as partial occlusions, head rotations, and varying lighting conditions. A slight decrease in accuracy was observed in low-resolution cameras, where finer facial details were harder to capture.

The system estimated head orientation using Euler angles (pitch, yaw, and roll), allowing it to determine head movements such as nodding, tilting, and turning. Head tracking was stable in well-lit environments but showed minor deviations in cases of poor lighting or sudden movements.

The system detected gaze direction accurately for focused and distracted states. It was able to track gaze deviations when subjects looked away from the screen. Inconsistent results appeared in cases of low camera resolution and reflections on glasses.

The system detected blinks by analyzing eyelid closure frequency. Normal blink rates were observed in attentive states, while higher or lower blink rates indicated drowsiness or distraction. Challenges included misclassification in cases of partial eyelid closure or rapid eye movements.

```
Epoch 47/50
448/448 ━━━━━━━━━━━━━━━━━━━━ 99s 220ms/step - accuracy: 0.6945 - loss: 0.8348 - val_accuracy: 0.6011 - val_loss: 1.0800
Epoch 48/50
448/448 ━━━━━━━━━━━━━━━━━━━━ 0s 177us/step - accuracy: 0.5938 - loss: 1.1030 - val_accuracy: 0.4000 - val_loss: 1.4192
Epoch 49/50
448/448 ━━━━━━━━━━━━━━━━━━━━ 92s 206ms/step - accuracy: 0.7048 - loss: 0.8065 - val_accuracy: 0.6024 - val_loss: 1.0764
Epoch 50/50
448/448 ━━━━━━━━━━━━━━━━━━━━ 0s 101us/step - accuracy: 0.7812 - loss: 0.6448 - val_accuracy: 0.6000 - val_loss: 0.9918
```

Figure 4.1: Training and validation accuracy of the CNN-based model

The results of the emotion detection model trained using a Convolutional Neural Network (CNN). The dataset consists of 28,709 training images and 7,178 validation images, categorized into 7 classes. Images are grayscale and resized to 48x48 pixels. The testing dataset contained 7,178 images, ensuring a balanced test set. Images were preprocessed by normalizing pixel values (dividing by 255) to scale them between 0 and 1, which helps improve convergence. The CNN was trained for 50 epochs as shown in Figure 4.1, with a batch size of 64. As shown in Figure 4.2, the metrics were visualized using line plots, the training accuracy improved progressively, reaching 78.12% by the final epoch. The training loss also decreased over epochs, which suggests that the model was learning effectively from the training data. The batch size was set to 64, which is generally a good choice for training CNNs efficiently without excessive memory usage. The training accuracy steadily improved across epochs, suggesting that the model was effectively learned from the dataset. The final training loss was 0.6448, meaning the model had a relatively low error when predicting on training data.

Figure 4.2: Model accuracy and loss of the CNN-based emotion detection model

4.2 Research Findings

In many educational settings, the impact of attention span on learning outcomes is often overlooked. Students who struggle with maintaining focus may face learning difficulties, but these issues frequently go undetected due to the lack of an objective evaluation system. Without recognizing this issue, students may continue to struggle academically, while educators may lack the necessary tools to intervene effectively. Therefore "DiverseMind" stands out addressing this issue as a web application to assess and monitor students' attention levels in real time. By detecting attention fluctuations, the system helps teachers identify students who require additional support, while also enabling students to gain awareness of their own focus patterns.

The system integrates Media pipe-based models, including facial landmark detection, head pose estimation, eye gaze tracking, and blink rate detection, to measure engagement. These techniques allow the system to analyze real-time visual behavior, making it easier to identify whether a student is highly focused or easily distracted. This data-driven approach provides valuable insights for educators, helping them adapt teaching methods to better accommodate individual learning needs.

Additionally, a CNN model was trained to enhance emotion detection, offering a deeper understanding of how students' emotional states impact their attention spans. The CNN model was trained on grayscale images resized to 48x48 pixels, ensuring efficiency while maintaining accuracy. With over 50 epochs of training, the model demonstrated strong learning capabilities, making it a robust choice for emotion classification. CNNs are particularly effective in image-based learning tasks due to their ability to capture spatial hierarchies in data, allowing for accurate recognition of subtle facial expressions that indicate engagement or distraction.

The training process was conducted in Jupyter Notebook, enabling efficient model experimentation, hyperparameter tuning, and performance visualization. Various data preprocessing and augmentation techniques were applied to improve model robustness, ensuring reliable performance across diverse conditions. The findings indicate that tracking visual engagement, cognitive load, and emotional state provides a comprehensive and reliable method for evaluating attention spans in students.

The results confirm that a short attention span correlates with learning difficulties, emphasizing the importance of real-time assessment tools in education. By integrating both MediaPipe models and CNN-based emotion detection, the system offers personalized insights, enabling educators to implement more effective interventions. Its scalability and adaptability make it a valuable tool for educational purposes.

4.3 Discussion

The integration of Media pipe-based models for attention evaluation, alongside emotion detection, has significantly enhanced the system's ability to analyze user engagement in real-time. The combination of facial landmark detection, head pose estimation, eye gaze tracking, blink rate detection, and emotion recognition provides a comprehensive approach to assessing attentiveness and emotional responses.

The emotion detection model effectively classifies facial expressions based on a dataset of grayscale images resized to 48x48 pixels, through CNN architecture. The model's ability to recognize emotions contributes to understanding user behavior, as emotions are closely linked to attention levels. By incorporating emotion recognition, the system can assess not only whether a user is focused but also their emotional state during engagement, offering deeper insights into cognitive processing and response patterns.

The Media pipe framework has proven to be a robust solution for tracking facial features with high precision. Facial landmark detection provides a strong foundation for extracting facial expressions and movements, supporting both emotion analysis and attentiveness evaluation. Head pose estimation allows the system to monitor the user's orientation and focus, determining whether they are actively engaged or shifting attention elsewhere. Eye gaze tracking further refines the attention assessment by detecting screen focus or gaze deviations, contributing to a more detailed evaluation of user engagement. Blink rate detection adds another dimension by capturing changes in cognitive load and alertness, as blinking patterns are closely associated with concentration levels.

By combining these components, the system presents a multi-model approach for monitoring attention, capable of operating efficiently in diverse environments. The ability to track multiple behavioral indicators simultaneously ensures real-time responsiveness.

To validate the system's effectiveness, testing was conducted at Munidasa Kumaratunga Vidyalaya Kaduwela, where students participated in controlled assessments of the system, designed to evaluate accuracy and reliability. Therefore, results highlight the potential of these models in advancing human-computer interaction and behavioral analytics, paving the way for further enhancements in personalized and adaptive attention-based systems.

5. CONCLUSION

In the Sri Lankan context, the research carried out on slow learners has been exceedingly limited, underscoring a substantial void in the existing scholarly and the practical understanding of the subject. To address this void “DiverseMind” a web-based solution is implemented in native Sinhala language for slow learners among primary school students. By integrating assessment with tailored interventions, the application provides a holistic framework for addressing learning challenges in writing, mathematics, attention span and memory. By employing advanced algorithms and animated user interfaces, it serves as a valuable resource for educators, parents and professionals in understanding and supporting unique needs of slow learners.

Attention span plays a crucial role in the learning process, yet limited research and support for slow learners in Sri Lanka have left a significant gap in addressing this challenge. “Diverse Mind”, developed in Sinhala, directly tackles this issue by providing an interactive platform that assesses and enhances attention span through engaging activities and gamified interventions. By identifying students' focus-related difficulties early, the application enables implement targeted strategies for improvement. By continuously refining its capabilities, “DiverseMind” has the potential to revolutionize the way attention span difficulties are addressed in young learners. The platform not only bridges existing gaps in educational support but also empowers slow learners with the tools they need to strengthen their focus, enhance their cognitive abilities, and unlock their full learning potential. Through sustained development and innovation, it paves the way for a more inclusive and effective learning experience for children who struggle with attention-related challenges.

Therefore, encouraged by the results and student engagement, C. A. D. Dhaya Nishanthi, a primary school in-service advisor from the Zonal Education Department of Jayawardhanapura, recommended and approved the framework for wider use, expressing interest in officially introducing it at the zonal level.

The future enhancement of this application provides vast promising benefits and opportunities. A mobile version for both Android and iOS platforms would enhance the accessibility and usability for a wider range of users. Expanding the language support beyond Sinhala language to cater diverse linguistic groups would extend its global reach. Accessibility enhancements for vision impaired and hearing-impaired uses would ensure its equitable usage for individuals with disabilities. Thus, pursuing these advancements, the application has the potential to develop into a holistic and impactful tool capable of addressing the diverse needs of learners and educators worldwide.

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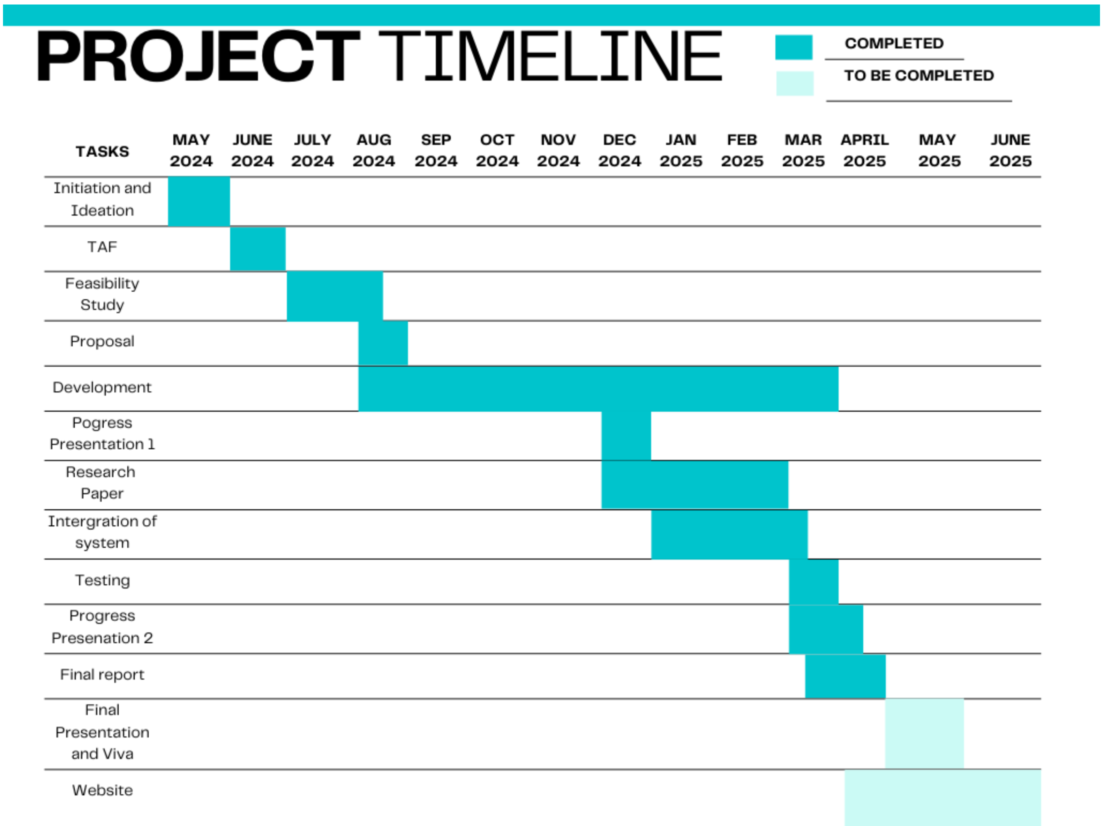
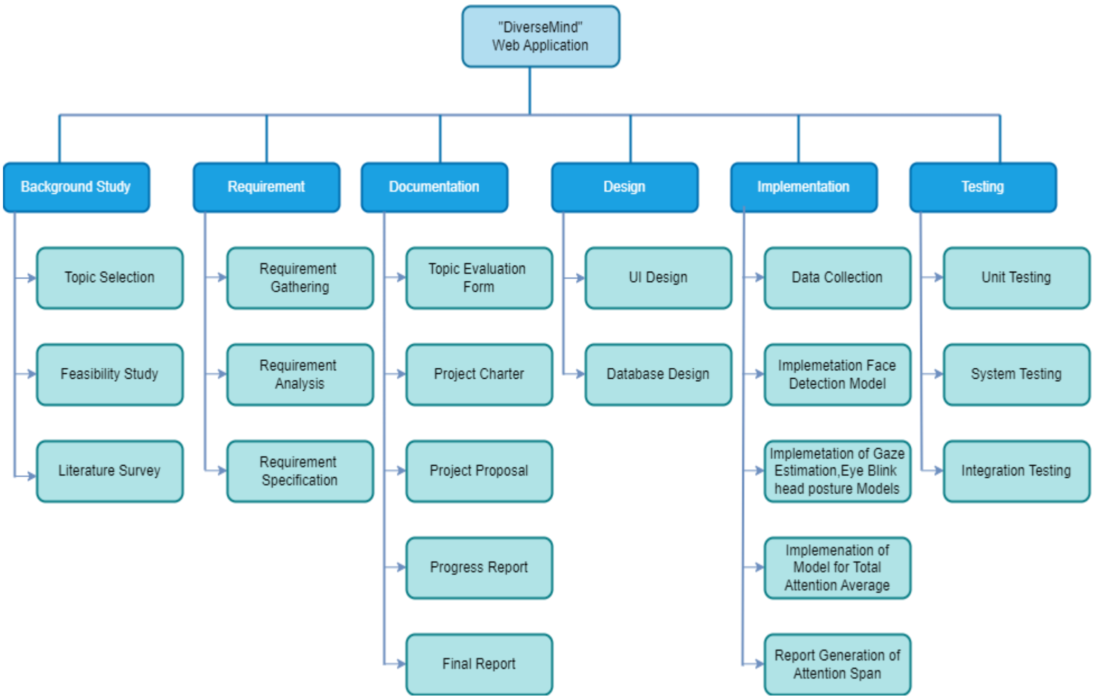
APPENDICES

Appendix A: Application Logo (DiverseMind)

Appendix B: Gantt Chart



Appendix C: Work Breakdown Chart



Appendix D: Recommendation and Testing Confirmation Letter from School Principal (Munidasa Kumarathunga Vidyalaya Kaduwela)

බප/ජය/මුනිදාස කුමාරතුංග විද්‍යාලය කඩුවෙල
WP/JAYA/ MUNIDASA KUMARATHUNGA VIDYALAYA KADUWELA.

දුරකථන Tel: } 011-2539232

දිනය Date } 2025.03.14

ඊමේල් E-mail } munidasakumarathunga@gmail.com

Vice principal,

Mrs.Ruwani Deepika,

Munidasa Kumarathunga Vidyalaya,

Kaduwela

15/03/2025

To Whom It May Concern,

Letter of Confirmation and Recommendation Regarding Student Testing Conducted for the 'DiverseMind' Application Research

I am writing to confirm that, Jayasundara J.M.H.H (IT21295492),Neewin S.L (IT21302930),Kiriwaththuduwa K.C.N (IT21298158), Herath H.M.R.M.K.(IT21217654), visited our school to conduct testing with Grade 4 students as part of their research on the "DiverseMind "application.

During their visit, they carried out assessments with selected students, gathering valuable insights into the effectiveness of their system. Based on the evaluations and observations, I recommend "DiverseMind" application as a research-level tool that has shown great potential in supporting slow learners. We found their approach to be highly structured, engaging, and beneficial for the students involved.

We appreciate their efforts and recognize that this application could be a valuable resource for both educators and students. I would like to recommend it to more schools for their usage and also wish them to develop the tool in further levels.

Thank you!

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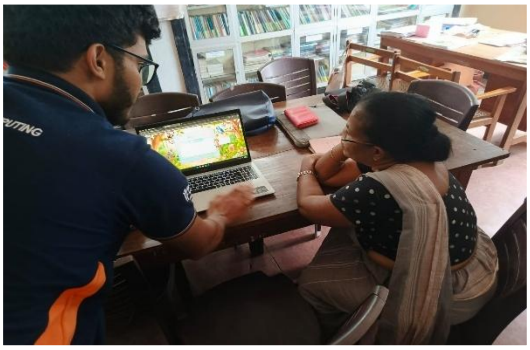
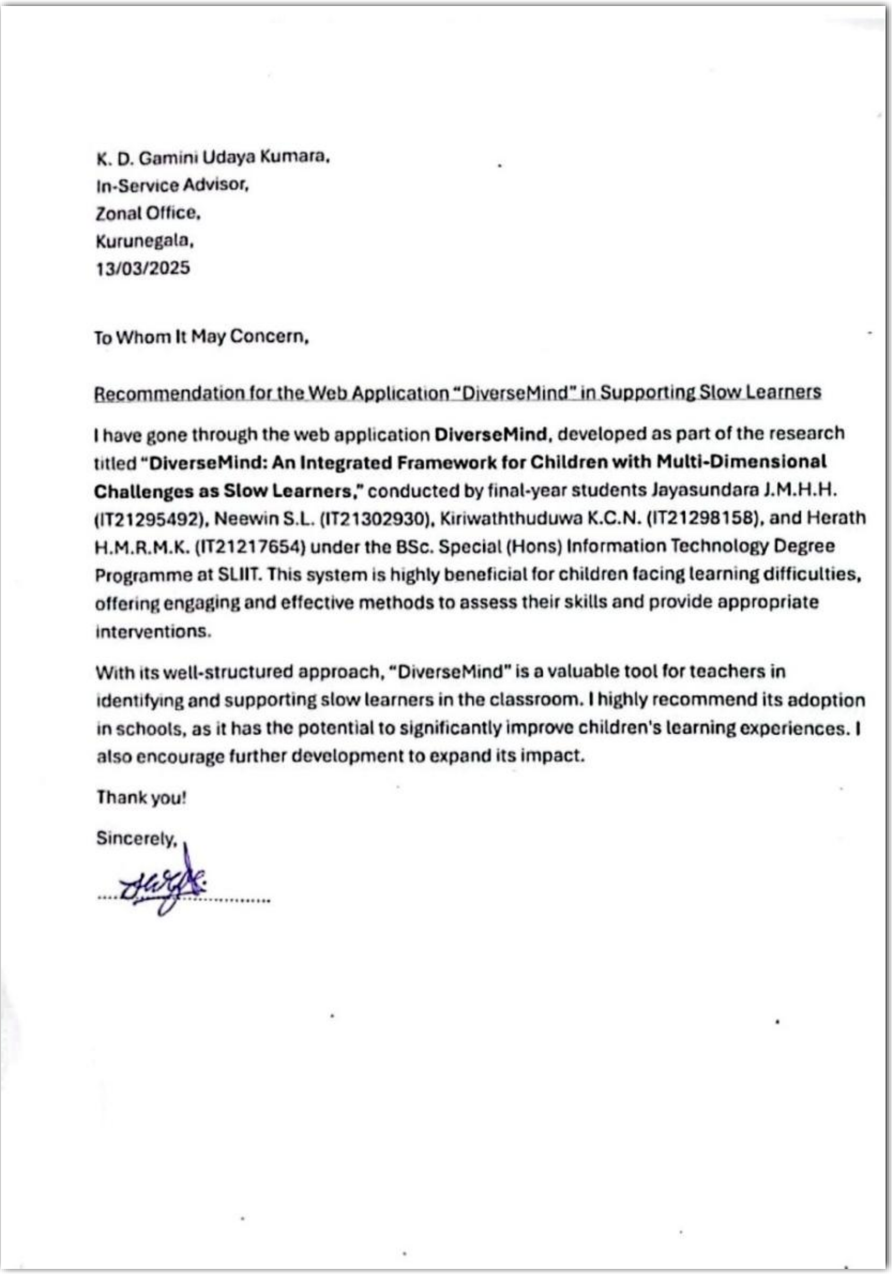
Appendix E: Photographic Evidence from School Visit for System Testing



Appendix F: Recommendation Letter from In-Service Advisor of Zonal Office Kurunegala



K. D. Gamini Udaya Kumara,
In-Service Advisor,
Zonal Office,
Kurunegala,
13/03/2025

To Whom It May Concern,

Recommendation for the Web Application "DiverseMind" in Supporting Slow Learners

I have gone through the web application **DiverseMind**, developed as part of the research titled **"DiverseMind: An Integrated Framework for Children with Multi-Dimensional Challenges as Slow Learners,"** conducted by final-year students Jayasundara J.M.H.H. (IT21295492), Neewin S.L. (IT21302930), Kiriwaththuduwa K.C.N. (IT21298158), and Herath H.M.R.M.K. (IT21217654) under the BSc. Special (Hons) Information Technology Degree Programme at SLIIT. This system is highly beneficial for children facing learning difficulties, offering engaging and effective methods to assess their skills and provide appropriate interventions.

With its well-structured approach, "DiverseMind" is a valuable tool for teachers in identifying and supporting slow learners in the classroom. I highly recommend its adoption in schools, as it has the potential to significantly improve children's learning experiences. I also encourage further development to expand its impact.

Thank you!

Sincerely,

Appendix G: Confirmation Letter for Data Collection (Munidasa Kumarathunga Vidyala Kaduwela)

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WP/JAYA/ MUNIDASA KUMARATHUNGA VIDYALAYA KADUWELA.

දුරකථන Tel: } 011-2539232

ඊමේල් E-mail } munidasakumarathunga@gmail.com

දිනය Date } 2025.03.14

Vice principal,
Mrs.Ruwani Deepika,
Munidasa Kumarathunga Vidyalaya,
Kaduwela
15/03/2025

To Whom It May Concern,

Confirmation of Successful Data Collection for "DiverseMind" Research Project

I hereby confirm that,

Jayasundara J.M.H.H (IT21295492), Neewin S.L (IT21302930),Kiriwaththuduwa K.C.N (IT21298158), Herath H.M.R.M.K.(IT21217654) group members of research project of "DiverseMind": screening and interventions to identify slow learners , have successfully completed the data collection from our school.

Thank you!

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Appendix H: Confirmation Letter for Data Collection (Sri Subhuthi National School Battaramulla)

බප/ජය ශ්‍රී සුභූති ජාතික පාසල, බත්තරමුල්ල

WP/JAYA SRI SUBHUTHI NATIONAL SCHOOL,

BATTARAMULLA

20.25/03/.12 දින

Dear Sir,

I hereby confirm that Jayasundara J.M.H.H (IT21295492), Neewin S.L (IT21302930), Kiriwaththuduwa K.C.N (IT21298158), and Herath H.M.R.M.K. (IT21217654), fourth-year undergraduates, who are conducting their final year research project titled "An Integrated Framework for Children with Multi-Dimensional Challenges as Slow Learners," have successfully completed the data collection at our school.

Principal

Chaturа Dammika Welaratne
Principal
S.L.P.S. - I
Sri Subhuthi National School
Battaramulla

Telephone : 011 2865945
Fax : 011 2879035
E-mail srisubhuthimmv@gmail.com

Appendix I: Photographic Evidence from School Visit for Data Collection






Appendix J: Confirmation Letter for the Appropriateness of the Designed System's Assessments (In-Service Advisor of Zonal Office Kurunegala)

K. D. Gamini Udaya Kumara,
In-Service Advisor,
Zonal Office,
Kurunegala,
09/11/2024

To Whom It May Concern,

Confirmation of System Appropriateness for Identifying Slow Learners

As part of the Bsc Special (Hons) Degree Programme at SLIIT, final-year students are required to complete a software project and a research-based dissertation. Under the research topic titled **"DiverseMind: An Integrated Framework for Children with Multi-Dimensional Challenges as Slow Learners,"** the following students have sought guidance and advice regarding the design of activities and data collection for their system:

- Jayasundara J.M.H.H. – IT21295492
- Neewin S.L. - IT21302930
- Kiriwaththuduwa K.C.N. - IT21298158
- Herath H.M.R.M.K. - IT21217654

This research project aims to identify slow learners among grade four primary school students by assessing their writing skills, mathematical abilities, attention span, and memory retention and provide proper inventions to improve their skills. The activities selected for the software are designed to align with the research objectives and are found to be appropriate for assessing the skills effectively. Based on my observations and the explanations provided by the students, I can confirm that the activities used in their system are suitable for identifying slow learners as intended.

Sincerely,

කේ.ඩී. ගාමිණී උදය කුමාර
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මාවතගම.

Appendix K: Confirmation Letter for Data Collection (Zonal Education Office Sri Jayewardenepura)

General Telephone No } 011-2870828

Zonal Director of Education } 011-2874077

Accountant } 011-2870820

Fax } 011-2874077

E-mail } sjpzone@gmail.com

කලාප අධ්‍යාපන කාර්යාලය
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Zonal Education Office
Sri Jayawardenapura
வலயக் கல்விக் காரியாலயம்
ஸ்ரீ ஜெயவர்த்தனபுரம்

My Number } බප/ජය/අප801/සමි.දත්ත

Your Number } ....

Zonal Education Office, Sri Jayawardanapura, Battaramulla.

Web: www.japurazone.sch.lk

Date : } 2024 . 08 . 14

විදුහල්පති,
කොතලාවල මහා විද්‍යාලය / යසෝදරා මහා විද්‍යාලය / මාලබේ ආදර්ශ පිරිමි විද්‍යාලය / රාහුල බාලිකා විද්‍යාලය / ලංකා සභා කනිෂ්ඨ විද්‍යාලය / සුබෝධි කනිෂ්ඨ විද්‍යාලය / සුහුනි ජාතික පාසල

<u>අවසාන වසර පර්යේෂණ ව්‍යාපෘතිය සඳහා දත්ත රැස් කිරීමට අවසරය සහ සහාය ඉල්ලා සිටීම</u>

උක්ත කරුණ සම්බන්ධයෙන් මාලබේ ශ්‍රී ලංකා තොරතුරු තාක්ෂණ ආයතනයේ තොරතුරු තාක්ෂණ විද්‍යාවේදී (ගෞරව) උපාධිය හදාරන ජේ.එම්.එච්.එච්. ජයසුන්දර මහතා ගේ 2024.08.14 දිනැති ඉල්ලීම් ලිපිය හා බැඳේ.

02. ඒ අනුව පහත නම සඳහන් සිව් දෙනෙකුගෙන් සමන්විත ශිෂ්‍ය කණ්ඩායම හට විශ්ව විද්‍යාලයීය උපාධිය ට අදාල පර්යේෂණ ව්‍යාපෘතිය සඳහා **"Diverse Mind: An Integrated Framework for Children with Multi-Dimensional Challenges as Slow Learners"** යන මාතෘකාව යටතේ අවශ්‍ය කරන දත්ත ලබා ගැනීමට විදුහල්පති ගේ එකඟතාවය මත පහත කොන්දේසි යටතේ මෙයින් අවසර ලබා දෙමි.

ශිෂ්‍ය නම	ශිෂ්‍ය ලියාපදිංචි අංකය	ජා. හැඳුනුම්පත් අංකය	සම්බන්ධතා අංකය
ජේ.එම්.එච්.එච්. ජයසුන්දර	IT21295492	200028202364	0769435437
එස්.එල්. නිවීන්	IT21302930	200171601984	0715177104
කේ.සී.එන්. කිරිවත්තුඩුව	IT21298158	200017502842	0701399308
එච්.එම්.ආර්.එම්.කේ. හේරත්	IT21217654	200167703099	0770870361

- විදුහල්පතිගේ අවසරය මත පමණක් ගුරු භවතුන්ගේ අධීක්ෂණය හා භාරකාරත්වය යටතේ පාසල් සිසුන් සහභාගි කරවා ගත යුතුය.
- ඉගෙනුම්, ඉගැන්වීම් හා ඇගයීම් ක්‍රියාවලියටත්, ළමුන්ගේ ආරක්ෂාවට, විනයට බාධාවක් නොවන පරිද්දෙන් හා පාසලේ ගෞරවයට හානි නොවන ලෙස පමණක් මෙම වැඩසටහනට ළමුන් සහභාගි කල යුතුය.
- දරුවන්ගේ ආරක්ෂාව පිළිබඳ සම්පූර්ණ වගකීම විදුහල්පති සතුවිය යුතුය.
- පාසලට ඇතුල් වීමේදී ඊට ගැලපෙන ඇඳුමකින් සැරසී සිටිය යුතුය.
- බෝවන රෝග පැතිරීම වැළැක්වීම සඳහා ගත යුතු ආරක්ෂක ක්‍රමවේද අනුගමනය කල යුතුය

ඩී.එන්.එස්.කේ. රණසිංහ
නියෝජ්‍ය කලාප අධ්‍යාපන අධ්‍යක්ෂ
(අධ්‍යාපන පරිපාලන)
කලාප අධ්‍යාපන කාර්යාලය
ශ්‍රී ජයවර්ධනපුර.

D.N.S.K. Ranasinghe
Deputy Zonal Director of Education
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පිටපත: 01. ජේ.එම්.එච්.එච්. ජයසුන්දර මහතා - අවශ්‍ය කටයුතු සඳහා

Appendix L: Survey Conducted on General Public

මන්දගාමී ඉගනුම් ක්‍රියාකාරකම් සහිත දරුවන් පිළිබඳ දැනුම සහ ඔවුන් හඳුනාගැනීමේ ප්‍රවේශයන් ඇගයීම (මෙම අධ්‍යනය ප්‍රථමික පාසල් දරුවන් කේන්ද්‍ර කරගන්ගනිමින් සිදුකර ඇත) | Evaluation of knowledge about children with slow learning and approaches to their identification (this study focused on primary school children)

මෙම සමීක්ෂණය අධ්‍යාපනඥයින්, දෙමාපියන්, භාරකරුවන් සහ සාමාන්‍ය ජනතාවගෙන් ප්‍රාථමික පාසල්වල මන්දගාමී ඉගෙන ගන්නන් පිළිබඳ ඔවුන්ගේ දැනුම සහ අත්දැකීම් පිළිබඳව අවබෝධයක් ලබා ගැනීමටයි. (The survey is conducted to gain insights from educators, parents, guardians and the general public about their knowledge and experiences of slow learners in primary schools.)

මන්දගාමී ඉගෙන ගන්නන්
(Slow Learners) යනු තම සම වයසේ මිතුරන් හා සසඳන විට නව සංකල්ප ග්‍රහණය කර ගැනීමට වැඩි කාලයක් හා පුහුණුවක් අවශ්‍ය වන දරුවන් ය. (Slow learners are children who need more time and practice to grasp new concepts compared to their peers.)

ප්‍රාථමික පාසල් දරුවන්ගේ මන්දගාමීව ඉගෙන ගන්නන් පිළිබඳ සංකල්පය ඔබ කෙතරම් හුරුපුරුදු ද? (How familiar are you with the concept of slow learners in primary schools?) *

	1	2	3	4	5	
හුරුපුරුදු නොවේ (Not familiar)	○	○	○	○	○	ඉතා හුරුපුරුදුය (Very familiar)

මන්දගාමී ඉගෙනුම් හැකියාවන් ඇති දරුවන්ගේ දුෂ්කරතා ඔබ අත්විඳ තිබේද හෝ සමීපතයෙකු තුලින් දැක තිබේ ද? (Have you experienced the difficulties of children with slow learning abilities or seen it in a close one ?) *

- ○ අත්දැකීමට ලක්ව තිබේ (Experienced)
- ○ අත්දැකීමට ලක්ව නැත (Not Experienced)
- ○ සමීපතයෙකු තුලින් දැක තිබේ (Seen it in a Close One)

ළමුන්ගේ මන්දගාමී ඉගෙනුම් හැකියාවන් කලින් හඳුනාගැනීමේ වැදගත්කම අගයන්න (Rate the importance of early detection of slow learning abilities in children) *

	1	2	3	4	5	
වැදගත් නොවේ (Not important)	○	○	○	○	○	ඉතා වැදගත් (Very Important)

පන්ති කාමරය තුළ මන්දගාමීව ඉගෙන ගන්නන් හඳුනා ගැනීමේදී සාම්ප්‍රදායික ප්‍රවේශයන් කෙතරම් ඵලදායී යැයි ඔබ සිතනවාද? (How effective do you think traditional approaches are in identifying slow learners with in the classroom?) *

	1	2	3	4	5	
ඵලදායී නොවේ (Not Effective)	○	○	○	○	○	ඉතා ඵලදායී (Very Effective)

සාම්ප්‍රදායික ක්‍රම හා සසඳන විට මන්දගාමීව ඉගෙන ගන්නන් හඳුනා ගැනීමට වෙබ් යෙදුමක් කෙතරම් ඵලදායී වනු ඇතැයි ඔබ සිතනවාද? (How effective do you think a web application would be in identifying slow learners compared to traditional methods?) *

	1	2	3	4	5	
ඵලදායී නොවේ (Not Effective)	○	○	○	○	○	ඉතා ඵලදායී (Very Effective)

මන්දගාමීව ඉගෙන ගන්නන් හඳුනා ගැනීම සහ ඔවුන්ගේ කුසලතා වැඩි දියුණු කිරීම සඳහා ඔබ වෙබ් යෙදුමක් නිර්දේශ කරන්නේද ? (Would you recommend a web application to identify slow learners and to improve their skills?) *

- ○ ඔව් (Yes)
- ○ නැත (No)

Submit

Clear form

Appendix M: Survey Conducted on Primary School Teachers

මන්දගාමී ඉගනුම් ක්‍රියාකාරකම් සහිත දරුවන් පිලිබඳ දැනුම සහ ඔවුන් හදුනාගැනීමේ ප්‍රවේශයන් ඇගයීම (මෙම අධ්‍යනය ප්‍රථමික පාසල් දරුවන් කේන්ද්‍ර කරගන්ගනිමින් සිදුකර ඇත) | Evaluation of knowledge about children with slow learning and approaches to their identification (this study focused on primary school children)

ඔබ ප්‍රාථමික පාසල් දරුවන් අතර මන්දගාමී ඉගෙනුම් සහිත දරුවන් හඳුනාගෙන තිබේද? (Have you identified slow learners among primary school children?)

- ඔව් (Yes)
- නැත (No)

ළමුන්ගේ මන්දගාමී ඉගෙනුම් හැකියාවන් කලින් හඳුනාගැනීම වැදගත් යැයි ඔබ සිතනවා ද? (Do you think early detection of slow learning abilities in children is important?)

- ඔව් (Yes)
- නැත (No)

එවැනි දරුවන් සඳහා අවශ්‍ය ක්‍රියාකාරකම් හඳුනාගෙන තිබේද? (Are activities identified for such children?)

- ඔව් (Yes)
- නැත (No)

ඒවා නව තාක්ෂණය හා සම්බන්ධ කිරීම කොතරම් උචිතද? (How appropriate is it to connect them with new technology?)

	1	2	3	4	5	
උචිත නොවේ (Not appropriate)	○	○	○	○	○	ඉතා උචිතයි (Very appropriate)

සාම්ප්‍රදායික ක්‍රම හා සසඳන විට මන්දගාමීව ඉගෙන ගන්නන් හඳුනා ගැනීමට වෙබ් යෙදුමක් ඵලදායී වනු ඇතැයි ඔබ සිතනවාද? (How effective do you think a web application would be in identifying slow learners compared to traditional methods?)

- ○ ඔව් (Yes)
- ○ නැත (No)

මන්දගාමීව ඉගෙන ගන්නන් හඳුනා ගැනීම සහ ඔවුන්ගේ කුසලතා වැඩි දියුණු කිරීම සඳහා ඔබ වෙබ් යෙදුමක් නිර්දේශ කරන්නේද ? (Would you recommend a web application to identify slow learners and to improve their skills?)

- ○ ඔව් (Yes)
- ○ නැත (No)

Submit Clear form

Appendix N: Plagiarism Report

24-25J-230_IT21302930.pdf

ORIGINALITY REPORT

9%	5%	5%	5%
SIMILARITY INDEX	INTERNET SOURCES	PUBLICATIONS	STUDENT PAPERS

PRIMARY SOURCES

1	Submitted to Sri Lanka Institute of Information Technology Student Paper	1 %
2	Submitted to National School of Business Management NSBM, Sri Lanka Student Paper	1 %
3	www.coursehero.com Internet Source	1 %
4	www.mdpi.com Internet Source	<1 %
5	hdl.handle.net Internet Source	<1 %
6	H.L. Gururaj, Francesco Flammini, S. Srividhya, M.L. Chayadevi, Sheba Selvam. "Computer Science Engineering", CRC Press, 2024 Publication	<1 %
7	Rahul RK, Shanthakumar S, Vykunth P, Sairamnath K. "Real-time Attention Span Tracking in Online Education", 2020 IEEE MIT	<1 %